Pooled powers and quadratic projections for ZFC consistency

Round 21 research report and reproducible artifacts

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Abstract

An explicit quartic in 217 unrestricted integers is given as the sum of 154 literal quadratic squares, with 7396 collected monomials and coefficient height 8880. It combines common-base power pooling with a strict-horizon factorial-free source, an odd beta multiplier, a wider remainder norm and certified integral projections. A 216-variable companion and a height-3200 companion retain different tradeoffs. The complete signed bootstrap is proved without assuming the radix or beta signs before they are recovered. Unit-homogenized equality cycles and shared coefficient factors strictly reduce the support of six matched coefficient-height-one quartics without adding variables. A separate ten-positive-variable construction combines linear trace, free quotient and inverse-Lucas correction, attaining exact degree 214 at 87 binary operations, with 232/85 and 240/82 companions. Two eleven-integer circuits retain degrees 298 and 300 at costs 146 and 144. Complete new ordinary arguments, literal arrays, coherent numerical payloads and independent finite checks are supplied. Their consistency interpretation composes with explicitly identified inherited calibrations; no consistency proof, global minimum or new end-to-end formal-kernel endpoint is claimed.

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1 Results and precise comparison

The principal deliverable is the fixed ordinary polynomial

$$\boxed{P_{217}(x_0,\dots,x_{216}) = \sum_{j=0}^{153} R_j(x_0,\dots,x_{216})^2 = 0, \qquad (x_0,\dots,x_{216}) \in \mathbb{Z}^{217}.} \tag{1}$$

Every residual is printed in Appendix A. The literal arrays and fully collected expansion contain no unspecified parameter, variable exponent, sequence operator or hidden existential coordinate. The polynomial has exact degree four, 7396 nonzero monomials and coefficient height 8880.

Theorem 1.1. *For the displayed principal polynomial, and for each separately specified companion on its stated domain D, the new ordinary witness arguments composed with the identified inherited arithmetic and syntactic calibration give*

$$\mathrm{ZFC} \vdash \left[\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

No consistency assumption is used to prove this equivalence.

The assertion is not a proof of either consistency or unsolvability. The report separates the new unbounded arguments, exact finite algebra, and the retained logical interfaces. It does not claim a new complete proof-assistant kernel endpoint or a global minimum.

1.1 Direct quartics with moderate literal coefficients

Equation over unrestricted integers	Variables	Rows	Terms	Height
Round 20 balanced	238	175	5236	3968
Incoming Strict Horizons principal	236	173	4876	7968
Incoming Pooled Powers	225	162	7867	128446
New principal	217	154	7396	8880
New lower-height companion	217	154	14793	3200
New dimension companion	216	153	25602	5632

All entries have exact degree four. The new principal improves every listed measure against the strongest new pooled input: eight fewer variables and rows, 471 fewer terms and lower coefficient height. Against its actual factorial-free Strict Horizons source it saves 19 variables and rows, but increases support and height. The separate lower-height and dimension companions express real tradeoffs. The earlier Round 20 balanced point also retains its sparse, low-height advantage. There is no single definition-independent meaning of simplest.

The pooled input supplies the simultaneous exact-power module; it is not the arithmetic ancestor of the new principal. The latter's source chain runs through Strict Horizons and its factorial-free C222 construction. The emitted source edits, common-base groups, radical projections and inverse coordinate maps establish the composition. A coordinate count obtained merely by adding savings from unrelated packets would not.

1.2 Height one and compact scalar equations

At coefficient height one, unit-homogenized equality cycles and shared coefficient factors reduce support without changing the six selected variable counts. The 505-variable point goes from 20491 to 20405 terms; the 503-variable alternative goes from 24193 to 24183. The complete six-point table and the zero-set proof appear in Section 4.

The ten-positive-variable branch supplies three separately fixed circuits: exact degree/cost 240/82, 232/85 and 214/87. The degree-214 equation combines a free quotient with inverse-Lucas correction, linear trace and an unshifted auxiliary representative. The sign proof is essential: a

positive coordinate chart cannot be changed by an arbitrary integer shear. The unchanged large coefficient tuple is supplied in full.

Two eleven-unrestricted-integer circuits retain exact degrees 298 and 300 while reducing their costs to 146 and 144 binary gates. These use a different complete coefficient tuple. No ten-positive result is stated as an eleven-integer result or vice versa. Scalar support and expanded coefficient height are not claimed when not materialized.

2 Logical scope and literal polynomial definitions

2.1 What the consistency interpretation asserts

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes the unrestricted integers. The domain is part of an equation, not a presentation detail. An equation over $\mathbb{N}$ is not thereby an equation over $\mathbb{Z}$, and an unrestricted construction requires its own sign argument or an explicit domain conversion.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected endpoint P on the stated domain D, the claim is

$$\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{2}$$

The ordinary transformations proved here are composed with the identified inherited calibration. On that proof chain, formalization in ZFC yields

$$\mathrm{ZFC} \vdash [\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{3}$$

This report does not assert that either sentence in (3) is true. In particular it does not prove consistency, decide solvability, establish a global minimum, or supply a new end-to-end proof-assistant kernel check or encoded object-language ZFC derivation.

For the direct families, the retained ordinary logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The first relates the fixed numerical calculus to a finite ZF refutation; the second is the retained ZF-to-ZFC refutation equivalence. The arithmetic front implements that calculus using finite syntax constructors, reciprocal CRT carriers, power or factorial certificates as appropriate, and range tests. The scalar branches retain their separately identified native/compiler and refutation calibrations at distinct coherent coefficient tuples. The eleven-integer branch additionally uses its signed-rank and interpolation reductions. These calibrated routes and tuples are not interchangeable.

2.2 The modular structure of the proof

The proof chain has four logically different layers.

1. The inherited calibration identifies a fixed finite arithmetic relation with syntactic refutation. Its provenance and exact numerical payloads are part of the source closure.

2. Unbounded arithmetic arguments prove both witness directions for the certificate relations. These include Pell classification, CRT choices, factorial fixed points, integer signs, and rank or interpolation lemmas.

3. Exact polynomial transformations replace one finite presentation by another. Integral substitutions, restored monic coordinates, graph identities and coefficient collection establish the claimed zero-set correspondence.

4. Finite independent checkers verify the actual emitted arrays and graphs, not merely an abstract instance of a theorem.

A hash certifies file identity, not a theorem; an exhaustive coefficient check certifies an identity, not an unbounded Pell assertion. Each layer is used only for what it establishes. The new sections give complete arguments for this round's transformations; the retained source closure supplies the inherited arithmetic and logical inputs at their stated boundary.

2.3 Explicit finite arrays define the direct equations

For a direct endpoint, the ordered `variables` array in `equation.json` fixes the independent coordinates $x_0, \dots, x_{n-1}$. A term encoded as $[c, [i_1, \dots, i_d]]$ denotes $c \prod_{j=1}^{d} x_{i_j}$; the empty index list denotes the constant c. Repeated indices are retained and give ordinary powers. The m residual rows $R_0, \dots, R_{m-1}$ have integer coefficients and degree at most two. The equation is exactly

$$P(\boldsymbol{x}) = \sum_{j=0}^{m-1} R_j(\boldsymbol{x})^2 = 0. \tag{4}$$

Every coefficient and every coordinate appears in a finite supplied array. The adjacent `expanded_polynomial.json` stores the fully collected polynomial, including its constant term. There is no factorial, variable exponent, division, binomial coefficient, bounded quantifier, or sequence symbol among its algebraic primitives. Such expressions occur only in the proof explaining its zeros.

For either domain $D \subseteq \mathbb{Z}$, equation (4) is equivalent to $R_j(\boldsymbol{x}) = 0$ for every j, because each square is nonnegative. Exact degree four is established by a collected nonzero quartic coefficient, not merely by the upper bound $\deg R_j \leq 2$. All independent coordinates are counted; named arithmetic abbreviations are not new unknowns.

The height-one endpoints are defined directly by their collected coefficient arrays. Their graph certificates prove the displayed energy identity and its zero-set correspondence. The independent checker reconstructs the energy and compares every collected coefficient. The height claim therefore concerns the final polynomial, not only its generating rows.

2.4 Arithmetic graphs define the scalar equations

The scalar graph leaves are the stated independent variables, ordinary integer constants, or fixed authentic numerical ports. The eleven-integer branch and the ten-positive-integer branch have different domains and different complete coefficient tuples. Each internal node is binary addition, subtraction, or multiplication. An acyclic graph with one designated output therefore denotes a unique element of an ordinary polynomial ring over $\mathbb{Z}$. Registers abbreviate polynomials; they do not introduce hidden existential variables. There is no division or radical gate.

The eleven-integer source34 tuple is `mixed-parity-unrounded-1064921`, at coefficient source commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its ports include $\beta = 2^{1064921} - 1$, cone_A $= 2^{1113}$, cone_B $= 2^{666104}$, and the complete literal K_0, M_C, M_F payloads. The retained M_F already includes its native-value-plus-β shift. It must not be shifted twice or replaced by a port from a positive-domain tuple. The payload files contain complete minimal unsigned little-endian integers and are bound by byte length, bit length and SHA-256; neither a hash nor an ellipsis is substituted for a coefficient. The finite inherited compiler recipe reconstructs those values.

For the ten-positive branch the four ports β, K_0, M_C, M_F instead come from the complete indexed BFS943/radix67 tuple, with $\beta = 2^{590069} - 1$ and largest port bit length 592884. Every independent coordinate is strictly positive. Its M_F shift and packing inequalities belong to that tuple alone. The four-port and six-port files are reconstructed and authenticated separately; no numerical leaf is transferred merely because two ports share a name.

2.5 What each measure counts

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$, write

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes a nonzero constant coefficient. The number of residual rows is distinct from support and from the number of operations. Scalar cost counts live binary graph gates, including squares and multiplication by fixed constants; copying a register and reading a leaf are free. Dead nodes do not count. Unless an expanded scalar polynomial is materialized, its support and total coefficient height are unmeasured, not zero.

A point dominates another only when it is no worse in every displayed measure and strictly better in at least one, with domain and degree held fixed where required. This is a finite comparison among specified presentations. Height one, smaller dimension, sparse support and low graph cost are different objectives, and no single global simplicity ordering is claimed.

3 Common base powers and exact quadratic reduction

The strongest new direct input is the *Pooled Powers* equation in 225 unrestricted integers, not Round 20's 237-variable point. The two-row power certificate and its simultaneous signed exactness theorem below are credited to that incoming package. The common-Hessian projection is credited to *Quadratic Projection 234*. This round composes their actual finite arrays, changes selected arithmetic budgets, and transfers the power pooling to a separately calibrated factorial-free source. Savings are measured on emitted arrays; counts from different sources are never added on paper.

3.1 The inherited two row certificate

For $|A| > 1$ let $(\chi_A(k), \psi_A(k))$, $k \in \mathbb{Z}$, be the canonical signed Pell pair. Its recurrence matrix is $\left(\begin{smallmatrix} A & A^2-1 \\ 1 & A \end{smallmatrix}\right)$, with initial pair $(1, 0)$. Its determinant is one, so all signed indices are defined. Given a nonzero base n, put

$$T_n = 2An - n^2 - 1.$$

Then $\gcd(n, T_n) = 1$, and

$$\chi_A(k) + (n - A)\psi_A(k) \equiv n^k \pmod{T_n} \qquad (k \in \mathbb{Z}). \tag{5}$$

Indeed $(n - A)^2 \equiv A^2 - 1 \pmod{T_n}$; evaluating the fundamental unit at $n - A$ gives n. Positive powers follow by multiplication, and negative powers by invertibility. The Pell functions are proof notation, not primitives in the final polynomial.

For r requests (k_i, m_i) at this base, retain the existing coordinates $X_i = \chi_A(k_i)$, $Y_i = \psi_A(k_i)$ and $Z_i = AY_i$. Introduce only two new integer coordinates K, L, and impose

$$K - 2\sum_i Y_i(m_i - X_i) - (2A - n)L = 0, \tag{6}$$

$$\sum_i \left((m_i - X_i + Z_i)^2 - Y_i^2\right) - nK + L = 0. \tag{7}$$

Every term has degree at most two. Neither T_n nor any displayed summation abbreviation introduces another independent coordinate.

Lemma 3.1. *The two rows imply*

$$\sum_i \left(m_i - X_i + (A - n)Y_i\right)^2 = T_n \left(L - \sum_i Y_i^2\right).$$

Conversely, when each error on the left is divisible by nonzero T_n, there are integers K, L satisfying both rows.

Proof. Set $D_i = m_i - X_i + AY_i$ and $H = \sum_i Y_i(m_i - X_i)$. Eliminating K gives $\sum_i (D_i^2 - Y_i^2) - 2nH = T_n L$. Expanding $\sum_i (D_i - nY_i)^2$ subtracts exactly $T_n \sum_i Y_i^2$, proving the identity. For the converse choose

$$L = \sum_i Y_i^2 + T_n^{-1} \sum_i (D_i - nY_i)^2, \qquad K = 2H + (2A - n)L.$$

The divisibility assumption makes L integral and direct substitution proves both rows. No sign of K or L is required. □

Theorem 3.2 (Simultaneous exactness from a square frame)**.** *Suppose $W \geq 258$, $1 \leq r \leq 15$, $n \neq 0$, $|A| > W^W$, and $n^2, k_i^2, m_i^2 < W$. If the pairs are the specified canonical signed Pell pairs, the two quadratic rows force $m_i = n^{k_i}$ for every i. For $|n| \geq 2$ they also force $k_i \geq 0$; for $n = \pm 1$ all signed indices are allowed. Conversely all these exact powers admit K, L.*

Proof. Put $b = \lfloor \sqrt{W - 1} \rfloor \geq 16$. Every absolute base, index and proposed output is at most b. Multiply the aggregate congruence by n^{2b}, using (5), to get

$$T_n \mid E, \qquad E = \sum_i (m_i n^b - n^{b+k_i})^2 \geq 0.$$

All exponents are nonnegative. Since $b \geq 2$, $|m_i n^b - n^{b+k_i}| \leq b^{b+1} + b^{2b} \leq 2b^{2b}$. Consequently

$$E \leq 4r b^{4b} < W^{2b+1} \leq W^W,$$

because $4r \leq 60 < W$, $b^2 < W$ and $2b + 1 \leq b^2 + 1 \leq W$. On the other hand $|T_n| \geq 2|A| - W > |A| > W^W$. Divisibility forces $E = 0$, so every summand vanishes. A nonnegative k_i gives the asserted power by cancelling n^b. If $k_i = -j < 0$, cancellation gives $m_i n^j = 1$; this excludes nonunit bases and gives the stated power at the two units. For the converse each exact error is divisible by T_n, so apply Lemma 3.1. □

The absolute-value argument includes negative radices and the unit case; assuming a positive radix before decoding its sign would not suffice. The actual square-frame ledger, canonical-pair classification and request ports are bound to the literal source by the finite checker. A different affine frame that only gives $|a| < W$ cannot silently be substituted for this square-frame hypothesis.

3.2 Exact integral projections along quadratic radicals

Write each residual as $Q_i = q_i + \ell_i + c_i$, with homogeneous quadratic part q_i, linear part ℓ_i, and integer Hessian H_i. For an integer vector v with $H_i v = 0$ for every i, exact expansion gives

$$Q_i(y + sv) = Q_i(y) + s\ell_i(v). \tag{8}$$

The mixed term is $sy^T H_i v = 0$ and the pure term vanishes since $2q_i(v) = v^T H_i v = 0$.

Theorem 3.3. *Suppose $v_p = 1$ and one slope $a_r = \ell_r(v)$ is ± 1. With $y_p = 0$, let*

$$R_i(y) = Q_i(y) - a_i a_r Q_r(y), \quad i \neq r, \qquad a_i = \ell_i(v).$$

These integer quadratics have a common zero set in bijection with that of all Q_i, with one fewer variable and residual. The lift is $x = y - a_r Q_r(y) v$, and the inverse is $y = x - x_p v$.

Proof. Put $s = -a_r Q_r(y)$. Equation (8) gives $Q_r(y + sv) = 0$ and $Q_i(y + sv) = R_i(y)$ for $i \neq r$. Conversely an old zero gives $y_p = 0$ and $Q_r(y) = -x_p a_r$, so the displayed lift restores precisely x_p. All other $R_i(y)$ vanish by the same formula. The two maps are inverse and integral; the new rows are constant combinations of restricted quadratics. No division condition remains. □

A primitive direction alone would not justify division by a nonunit slope. The actual certificates require the two literal unit conditions. These unrestricted-integer coordinate changes need not preserve a positive domain.

Lemma 3.4 (Affine consequence elimination)**.** *Suppose a nonzero constant integer combination $L = \sum_i \lambda_i Q_i$ has degree at most one and its primitive affine normalization has coefficient ± 1 on x_p. Solve it as $x_p = S(\widehat{x_p})$. Substitute in every row and omit any row r with $\lambda_r \neq 0$. The resulting quadratic system has an integer zero set in bijection with the original system.*

Proof. Every original zero satisfies $L = 0$, hence $x_p = S$ and every retained substituted row. Conversely, the substitution makes L identically zero. At a zero of the retained rows, $\lambda_r Q_r = 0$; the integers have no zero divisors, so $Q_r = 0$. Insertion of S and deletion of x_p are inverse. A unit λ_r is not needed; a unit solved variable coefficient after primitive normalization is essential. □

An elementary integer row operation $Q_i \leftarrow Q_i + kQ_j$, $i \neq j$, has inverse with $-k$. A triangular integral affine coordinate change has its explicit integral inverse. A monic quadratic definition can be eliminated only when its complete substitution retains the claimed degree. Each released certificate records these conditions and the exact index compaction. A sum of squares of the final rows is generally a *different* polynomial, with the same zero-existence statement.

3.3 The actual factorial free source and square frame

The selected source is the literal 239-coordinate Strict Horizons source, not the factorial-authenticating Pooled Powers source. Its semantic values are

$$t = 16 + h^2, \quad F = t^t, \quad B = \binom{F+t}{t}, \quad q = 4z + 1, \quad H = 2 + C + \sum_{j=1}^{8} A_j.$$

Here B is initially only a proposed binomial value. There are nine actual CRT representatives, $C, A_1, \ldots, A_8$, and a globally evaluated derived label $J(\boldsymbol{A}, C)$, which is not an additional independent column. The unchanged endpoints and index row are

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + [1 + (t+1)y]u_t, \qquad Bh_B = C + F + 1. \tag{9}$$

The separate request at the already positive base t will first recover $F = t^t$. No coordinate is authenticated as $t!$ in this branch.

Before using the signs of these arithmetic values, the retained squared semantic anchor and deterministic comparison-center proof give all fourteen canonical signed Pell pairs. The literal actual-atom frame is $W = 2 + \sum a^2$, with exactly 34 squares. Its atoms comprise $t, F + t, x, y + 1, F, B$; the nine representatives; N and the global derived label; $q, q + 1, BH$; and the fourteen remaining request outputs. Consequently every base, index and output of all seventeen requests has square below W, and t^2 gives $W \geq 258$. The frame provides $|A| > W^W$ independently of the later sign bootstrap. Stored coordinate offsets matter: for example the source coordinate named y stores $y + 1$, and one schema coordinate stores $A_a + 1$. The checker expands the actual affine atoms, not their historical names.

Apply Theorem 3.2 to eleven requests at $q + 1$ and three at q, leaving the requests at t, B, BH separate. Each group replaces r private quotients and one modulus by two coordinates, saving $r - 1$ variables and rows: ten plus two. The retained AY_i wires make every new row quadratic.

3.4 An odd beta multiplier and a wider sphere

Replace the old $y = F(2v + 1)^2$ by

$$y = F(2v + 1). \tag{10}$$

The multiplier's defining row becomes affine and permits its monic elimination. This is a semantic extension followed by an exact projection, not a claim that every new multiplier is an old square.

For the eleven normalized remainders retain the constant one and replace coefficient four by two:

$$M^2 = 1 + 2\sum_{j=1}^{11} \rho_j^2 + G(a, b, c), \qquad G(a, b, c) = a^2 + b^2 + c(c + 1). \tag{11}$$

For integers $G \geq 0$, so $M \neq 0$ and $|\rho_j| < |M|/\sqrt{2} < 3|M|/4$. The constant one is essential to this nonzeroness argument and is not removed. The old signed-factorial proof of nonzeroness is not imported into this factorial-free source.

For a normalized row $Dv = M$, $vU = MQ + \rho$, the nonzero denominator and M give $v \neq 0$. The row gives $v \mid \rho$, so $r = \rho/v$ is integral and

$$U = DQ + r, \qquad |r| < 3|D|/4. \tag{12}$$

No sign of D, v, M or ρ was assumed.

3.5 Recover the signs in an acyclic order

The unpooled t request first gives $F = t^t > 2^{32}$. Equation (10) implies $|y| \geq F > 1$, but does *not* yet imply $y > 0$. The affine radix has $q \neq 0$ and $|q + 1| \geq 2$. Apply the $q + 1$ pool: all nine actual representatives and the derived-label index are nonnegative and all their numerators are exact. Hence $H \geq 2$ and the index row makes $B \neq 0$. Now $|BH| \geq 2$, so its separate request gives $x \geq 0$ and $q = (BH)^x$.

If $x = 0$, the last endpoint in (9) would require $1 + (t + 1)y$ to divide -1. Its absolute value is at least $(t + 1)F - 1 > 1$, impossible. Thus $x \geq 1$ and $|q| \geq 2$. Only now use the q pool to recover exact q^t, q^{y+1}, q^x and nonnegative indices. In particular $y \geq -1$; together with $|y| \geq F > 1$, this gives $y \geq F > 0$. This is the first justified use of the positive sign of y. The ratio $q^t v = q^x$ also gives $x \geq t$.

If $u_0 \leq -1$, the first endpoint would give $x \leq F - y \leq 0$. Therefore $u_0 \geq 0$, $x \geq F + 1 > 1$, $u_t \geq 1$ and

$$x \geq 2 + (t + 1)y \geq 2 + (t + 1)F. \tag{13}$$

Writing $Q = |q|$ and $b = |B|$, we have $Q = |BH|^x \geq 2^x$ and $b < Q/2$. No positivity of B or q was used to obtain these bounds.

3.6 Decode the binomial at a signed radix

Let $C_j = \binom{F+t}{j}$. For $0 \leq j \leq t$,

$$C_j \leq (F + t)^t \leq (2F)^t \leq 2^{t(F-1)}.$$

Here $t \leq F$ and $2F \leq 2^{F-1}$ for $F \geq 4$. Equation (13) implies $C_t < Q/32$ and $\max_{j \leq t} C_j/(Q-1) < 1/32$. The signed low tail $r_0 = \sum_{j<t} C_j q^j$ therefore satisfies $|r_0| < Q^t/32$, even for negative q. The restored carrier-digit row is

$$(q + 1)^{F+t} = q^t(B + qh_{\text{high}}) + r.$$

Thus $q^t \mid r - r_0$, while $|r - r_0| < (3/4 + 1/32)Q^t < Q^t$ by (12). Hence $r = r_0$. Subtract and divide by q^t to obtain $B \equiv C_t \pmod{q}$. Since $|B - C_t| < Q/2 + Q/32 < Q$, we conclude

$$B = \binom{F + t}{t} > 0, \qquad q = (BH)^x > 0. \tag{14}$$

These signs are consequences of the decoded digit, not its premises.

For every other genuine low digit, the retained source estimate is $0 \leq r_0 < D/32$. The same divisible-difference proof with radius $3D/4$ gives $r = r_0$. Thus the old strict interval witnesses $0 \leq r < D$ can all be restored, not merely the necessary congruences. Ordinary upper representatives have binomial coefficients at the required positions at most H^x. The global derived label has bound $J(\boldsymbol{A}, C) \leq H^{2^{22}}$, with $2^{22}t < x$. Since $q = B^x H^x$, these are the retained margins giving the $1/32$ tails. The source's local majorant uses exponent 2^{23}; $2 + t + x + y + F < 2x$ and $x > 2^{32}$ give

$$(2 + t + x + y + F)^{2^{23}} < (2x)^{2^{23}} < 2^x.$$

The last inequality follows from monotonicity of $x/\log_2(2x)$ above 2^{32} and its value there.

3.7 The retained prime power and flow interfaces

We record why the now restored source conditions suffice. Let $\Lambda_t = \mathrm{lcm}(1, \ldots, t)$. The elementary bound $\Lambda_t \le 4^{t-1}$ follows by induction: for $k = \lceil t/2 \rceil$, each additional prime-power factor in Λ_t/Λ_k divides the central binomial coefficient, which is at most 2^{t-1}. Therefore $\Lambda_t^2 \le 16^{t-1} < t^t = F$ for $t \ge 16$. This licenses the linear CRT radius F, without a factorial coordinate.

Two different prime-power facts are needed. First, if $p^e \| \binom{F+t}{t}$, then $p^e \mid F + i$ for some $1 \le i \le t$. Indeed the factorial valuation is a sum of zero-one terms at prime-power levels, and is at most $\max_i v_p(F + i)$; beyond that maximum no level contributes. Second, every row i has such a full prime power $p^e > t$. To see this, $F + i > \Lambda_t^2$ gives a prime with $v_p(F + i) > 2v_p(\Lambda_t)$. All other interval terms have $v_p(F + j) = v_p(j - i)$. The loss on division by $t!$ is

$$v_p\left(t\binom{t-1}{i-1}\right) \le v_p(\Lambda_t),$$

since each Legendre summand is zero or one and vanishes above t. The remaining full exponent is larger than $v_p(\Lambda_t)$, so $p^e > t$ and still divides $F + i$.

For $A \in \mathbb{N}$, divisibility $p^e \mid \binom{A}{k}$ forces $A \bmod p^e < k$. If $A \ge k$ and its falling interval contained no multiple of p^e, all valuation levels from e onward would vanish; the lower zero-one levels sum to at most $e - 1$, a contradiction. If $A < k$, the conclusion is immediate. A row's large selector decodes its bounded local tuple; the restored strict ranges identify the row, and the local bound $0 \le \mathcal{L} < 2^x \le p^{xe}$ makes its divisible local error zero. The derived label is decoded by polynomial congruence from the same tuple, not treated as an independent column. For the converse CRT construction, the every-prime fact assigns a compatible valid row at *every* prime of B. Precision p^{2xe} loses fewer than $x \le xe$ factors on division by each relevant factorial, leaving the required p^{xe} divisibility. One of these two prime-power lemmas alone would not establish both witness directions.

If $y = F$, the initial beta accumulator is zero. Positive-weight local updates propagate zero, contradicting the final accumulator one. Thus $y > F$, and the restored updates give

$$\prod_{i<t}(1 + Fv_i) = (1 + F)\prod_{i<t}(1 + Fp_i)(1 + Fq_i).$$

Discard zero labels. Unequal factor counts would have a smaller count $r \le t$ and force $(F+1)^{r+1} \le t^r(F + 1)^r$, impossible since $F + 1 > t^t$. Hence both products have the same number $r \le t$ of positive factors. In their product polynomials every coefficient of positive degree lies in $[1, F]$, so coefficient differences have absolute value below F. Reduction modulo F at the first nonzero difference coefficient shows that equality at F implies polynomial equality. Distinct factors $1 + jz$ then recover exactly the required label multiplicities. An individual coefficient is allowed to equal F; the strict bound is on its difference.

Finally, take the finite Horn closure of truth. If falsity were outside, every listed rule with an outside conclusion would have an outside premise. Summing conclusions minus premises over outside labels would be at most zero; the recovered multiplicities make it one. This contradiction gives a C222 refutation. The identified C222/ZF/ZFC interfaces in Section 2 complete soundness.

3.8 Constructive completeness and retained choice order

Begin with a finite calibrated C222 refutation. The unchanged Strict Horizons construction chooses $t = 16 + h^2$ strictly beyond its finite tree length, local parameters and derived labels, with h even. It then forms $F = t^t$, the finite pending-accumulator path and an old spacing $y = F(2v_{\mathrm{old}} + 1)^2$ large enough for that path. The even choice gives $v_2(F) = t\, v_2(t) > v_2(t!)$; a sufficiently large odd multiple of the odd part of $t!$ supplies the old multiplier. The beta moduli are pairwise coprime and odd, so their CRT code x can be chosen even and large. The retained binomial CRT at $N^2 = B^{2x}$ gives all representatives. The binomial B is odd because $v_2(F + i) = v_2(i)$ for

$i \le t$. Thus parity-adjustable representatives supply an even dominant derived anchor and odd H. Consequently $(BH)^x \equiv 1 \pmod 4$, which supplies the new affine radix coordinate. Only after the arithmetic atoms are fixed are fresh simultaneous canonical Pell/frame witnesses selected.

Keep all these old source values and set

$$v_{\text{new}} = \left((2v_{\text{old}} + 1)^2 - 1\right)/2 \in \mathbb{Z}.$$

This preserves actual y and every frame atom under (10). The true normalized remainders satisfy $|\rho_j| < M/32$, so $2\sum_j \rho_j^2 < 22M^2/1024 < M^2$. The positive integer $M^2 - 2\sum \rho_j^2$ is at least one, therefore $M^2 - 1 - 2\sum \rho_j^2 \ge 0$. Every nonnegative integer g is represented by G: the three-square theorem applied to $4g + 1$ gives exactly one odd square modulo four, which yields $g = a^2 + b^2 + c(c+1)$ on division by four. This supplies the wider sphere. Lemma 3.1 supplies the four pooled coordinates. All new rows now vanish. This is a constructive existence equivalence, not a bijection between arbitrary old and new beta multipliers or ternary-guard triples.

3.9 The exact algebra after the semantic changes

The private-wire reduction first stores $J_0 = s_0^2 + g^2$ instead of two coordinates used only through their sum. Restoration gives the two original squares. It next stores $J = FS$, retaining $K = FS'$, instead of three private factors $1 + Fb$, $1 + F(a-1)$ and $1 + Fd$, where $d = 6a + 8b + 2p$. The consumer identities are

$$S'(1 + Fb) = S' + Kb, \quad S(1 + F(a-1)) = S + J(a-1),$$

$$S(1 + Fb) - S'(1 + Fd) = S + Jb - S' - Kd.$$

Every old factor restores by its defining polynomial; the certificate checks the residual differences as multiples of the retained monic product definitions. The sum and product replacements save three variables and rows without creating cubic residuals.

Next put $D_i = X_i - Z_i$, with inverse $X_i = D_i + Z_i$. Subtract the fourteen retained norm rows from the orientation row. The pooled second row now uses $(m_i - D_i)^2 - Y_i^2$. These invertible shears and row operations change density, not dimension. Every transported Round 20 core operation is rechecked on this actual source: either an integral shear, a reversible row operation, or a monic projection with an integral degree-at-most-two restoration. A quadratic restoration is allowed only when every other occurrence of the deleted coordinate is linear. The new low-height core exposes two further common-Hessian directions, handled by Theorem 3.3. The dimension companion uses one additional affine consequence as in Lemma 3.4. Finally frozen integer row-basis operations select the support/height tradeoffs. No operation is licensed merely because its old coordinate name occurs in an earlier certificate.

3.10 Literal selected endpoints

The release paths `signed/principal217`, `signed/low217` and `signed/dimension216` contain the respective complete residual arrays, fully collected expansions and metrics. Their certificates replay the source edits and every algebraic operation; restores compose in reverse order. The principal's 154 residuals are printed in full in the appendix. Independent coefficient collection establishes exact degree four, coordinate liveness, support and height. The new arithmetic soundness and completeness, followed by these finite zero-set equivalences, prove Theorem 1.1 for all three direct outputs.

4 Unit homogenized equality at coefficient height one

The source compiler acts on a finite quadratic system over unrestricted integers. The inherited Round 20 construction already supplies certified normalization, opposite-occurrence fusion, polarization, anchored equality and internal-flow contraction. We give its complete energy and witness

argument to establish the interface, then prove this round's unit-homogenized cycle replacement. Every height claim concerns the entire collected polynomial, including all equality penalties.

4.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_cX_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_is_{c,j}$ otherwise. The cover condition proves that one option always exists. The retained deterministic greedy cover starts with the loop vertices and repeatedly selects a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and t_cX_i, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus normalization preserves both witness maps. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j : m_{rj} = m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

4.2 Share a coefficient across both monomial factors

The two refined compact profiles use a further exact normalization step. For a quadratic occurrence cxy, choose positive integers a, b with $|c| = ab$. Let $S(1, x) = x$; for $d > 1$ introduce a shared coordinate $S(d, x)$ with monic definition $S(d, x) - K_dx = 0$, where $K_d = d$ is fixed by the retained triangular constant chain. Replace the occurrence by

$$\operatorname{sgn}(c) S(a, x) S(b, y).$$

It remains a degree-at-most-two monomial of coefficient ± 1. Its unique constructor substitution is exactly cxy. If $x = y$ the same scale coordinate may occur twice, giving its ordinary square. Repeated definitions are shared; no unused scale coordinate is introduced.

This is an unconditional integral source equivalence. Given a source tuple, its monic constants and scales have unique values and every normalized row pulls back to its original row. Conversely those monic rows force the values and recover every original equation. There is no variable division or hidden divisibility assumption. The positive integer factor choices are frozen certificate data, not equation parameters.

The finite selection enumerates divisor pairs in both orientations and accepts a reduction of the number of used scales plus directly needed constants, then builds the actual constant chain. That cost is only a search heuristic. Exact final source pullbacks and coefficient expansion, not a presumed optimum of the heuristic, certify the delivered counts. The changed normalization alters the eventual graph namespace; its bijection with the old graph goes through the common arithmetic source, not an identity on arbitrary final tuples. The graph-energy and universal lift proof below applies unchanged to either normalized source.

4.3 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v \in B_r} b_v = T_r(y), \qquad \sum_{v \in B_0} b_v = -\sum_{r} T_r(y). \tag{15}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (15) is unchanged by fusion. This is the inherited occurrence-boundary identity.

4.4 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2} \sum_{v} d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - ab^3 = (a - b)^2 (a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell} (u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{16}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The retained square-alias transformation below preserves this energy identity and revises a selected equality penalty with its exact zero set intact.

4.5 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (15) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2} \left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x) \right)^2 \right). \tag{17}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (17) composes unchanged through affine refactorization. One must not replace (17) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (16) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all

normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

4.6 Square polarization removes equality dummies

Suppose a logical value currently has exactly two factor aliases a, b, and its equality penalty uses a fresh dummy d:

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

The reason for retaining this dummy can be a prescribed square a^2 inside the incidence graph: directly substituting the two-alias penalty $Q(a, b)$ may collide with its degree-four diagonal contribution. The retained Round 19 operation first changes selected prescribed squares into products of same-logical aliases, $a^2 \mapsto ab$ (or $b^2 \mapsto ab$). It then replaces $C(a, b, d)$ by $Q(a, b)$ and deletes the dummy coordinate d.

Theorem 4.1. *Make any finite selection of such same-logical prescribed-square changes, and replace each affected two-alias equality cycle by Q. If the full collected final polynomial has coefficient height one, it is a height-one quartic with an integer zero set bijective to the original arithmetic source. Each deleted equality dummy reduces its dimension by one.*

Proof. Changing a prescribed edge's label from a^2 to ab leaves its two incidence endpoints and orientation unchanged. The graph energy therefore remains exactly $\frac{1}{2}\sum_v d_v^2$, with its divergences evaluated using the modified labels. It is still nonnegative, and all labels still have degree at most two. Also

$$Q(a, b) = (a - b)^2(a^2 + ab + b^2) \geq 0,$$

and $Q(a, b) = 0$ if and only if $a = b$: the second factor equals $(a + b/2)^2 + 3b^2/4$ and vanishes only at $a = b = 0$.

At any zero of the final energy, every nonnegative equality term is zero, so same-logical aliases agree and units equal one. Consequently every polarized label ab has exactly the value previously specified by a^2. The occurrence-boundary identity is unchanged after logical substitution. Vanishing vertex divergences therefore give all normalized source rows, and monic reconstruction gives an actual arithmetic source zero.

Conversely, for an original source zero assign both a and b their common logical value. All polarized edge labels then have their old values, all new Q terms vanish, and the same unique subtree flow assignment makes every vertex divergence zero. Deleted dummies have unique old value $d = a = b$, so extension and deletion are inverse on the zero sets. This also proves that the source-polynomial lift identity (17) survives universally, not only at source zeros.

The degree is at most four by construction. Exact degree four and coefficient height one are finite properties of the entire collected polynomial and are explicitly checked. They do not follow merely from the simplicity of the incidence graph after monomial substitution. The operation is accepted only after this full collection check, including all equality and unit terms. Thus possible collisions do not invalidate the claim. □

The theorem makes no assertion that an arbitrary polarization is safe for height one, nor that a chosen finite sequence is optimal. It separates the unconditional witness proof from the coefficient test selecting a particular literal endpoint.

4.7 Absorbing equality dummies into anchored products

The retained anchored operation removes an equality dummy without changing any prescribed square label. Suppose two aliases a, b of one non-unit logical value are tied by a fresh dummy d through

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

Assume d occurs nowhere else. Let u be an existing unit alias already anchored to one by U. Substitute $d = au$ and delete the coordinate d. The new equality penalty is

$$\widetilde{C}(a, b, u) = a^2 + b^2 + a^2u^2 - ab - abu - a^2u. \tag{18}$$

It has degree four and is the literal polynomial $C(a, b, au)$.

Theorem 4.2. *Under the stated incidence restrictions, replacing $C(a, b, d)$ by* (18) *preserves the integer zero set by a polynomial bijection after deleting one coordinate. It also preserves the universal source lift identity* (17)*. If the entire resulting collected polynomial has height one, exact degree four and all declared coordinates present, it is a certified height-one quartic with one fewer independent integer variable.*

Proof. For arbitrary real a, b, u,

$$2\widetilde{C} = (a - b)^2 + (b - au)^2 + (au - a)^2 \geq 0.$$

At a zero of the complete energy, its unchanged unit penalty forces $u = 1$. The new equality penalty then vanishes exactly when $a = b$. Thus the old dummy has the unique value $d = a = b$, while every other summand is unchanged. Extending with $d = a$ maps a new zero to an old zero. Conversely, at an old zero $d = a = b$ and $u = 1$, so deleting d gives a new zero. These maps are inverse polynomial maps on the zero sets.

For a canonical source lift, all unit aliases equal one and all ordinary aliases equal their logical values. Consequently $au = a = d$ identically on that lift, including away from source zeros. No prescribed incidence label or divergence has changed, so the universal lift identity is retained.

Polynomial substitution preserves nonnegativity but not automatically coefficient height. The term a^2u^2 may collide with graph-energy terms, and several absorbed dummies may share a unit alias. The acceptance test therefore collects the whole energy after every proposed substitution, including all other penalties. It explicitly checks the finite claims about coefficients, degree and the live coordinate set. No extra unit coordinate is introduced: u was already present and anchored. □

Several such absorptions can be composed, using the same or different existing unit aliases, provided the full polynomial test accepts the frozen final choices. Unlike Round 19 polarization, this operation leaves the prescribed square monomials in the incidence graph untouched. Unlike a formal substitution $d = a$ alone, it does not insert a coefficient -2 into the equality row before the complete energy is considered. Its quartic product with an already anchored unit provides additional distinct monomials that can retain height one.

4.8 Contracting an internal flow edge

The retained operation changes the divergence energy rather than only the alias penalties. It removes a flow coordinate on an internal tree edge and identifies its endpoints. It is therefore not an alias identification or an instance of square polarization. Its soundness is unconditional; coefficient height one is a separate exact acceptance test.

Theorem 4.3. *Let an incidence presentation have the energy form* (16)*, with every internal edge belonging to a tree inside one row block or the hub block. Choose an internal edge from u to v, with its own flow coordinate w. Delete that edge, identify its vertices, and replace their two divergences*

by their sum. Retain all prescribed occurrence labels and equality and unit penalties. Then the old and new integer zero sets are in polynomial bijection, preserving the original arithmetic coordinates. The new presentation has one fewer flow coordinate and degree at most four. If full collection of its polynomial has height one, exact degree four, and every listed variable occurs, it is a certified height-one quartic on that smaller coordinate set.

Proof. Write the old two divergences as

$$d_u = a_u + \epsilon w, \qquad d_v = a_v - \epsilon w, \qquad \epsilon \in \{-1, 1\},$$

where a_u, a_v are integer polynomials independent of w. No other divergence or penalty depends on w, because it is the independent flow coordinate of this one edge. The merged divergence is $d_{uv} = a_u + a_v$. Thus the new polynomial is exactly

$$P' = \frac{1}{2}\left((a_u + a_v)^2 + \sum_{t \neq u,v} d_t^2\right) + \sum C + \sum Q + U.$$

This expression proves nonnegativity and the degree bound without any claim that the old and new polynomials themselves agree.

At an old zero, nonnegativity forces $d_u = d_v = 0$ and every other summand to vanish. Deleting w therefore gives $a_u + a_v = 0$ and a new zero. Conversely, at a new zero all its displayed summands vanish. Set

$$w = \epsilon a_v.$$

Then $d_v = 0$, while $d_u = a_u + a_v = 0$; all other old summands are unchanged. Indeed the stronger identity $P|_{w=\epsilon a_v} = P'$ holds for every remaining tuple: the two old divergences become $a_u + a_v$ and zero. This is an integral polynomial restoration, and it is unique on zeros because ϵ is a unit. Restoration and deletion are mutually inverse on the zero sets. All equality and unit arguments, and hence all recovered logical coordinates, are identical on the two sides.

The edge's endpoints belong to one block. Summing the contracted vertex divergences in that block therefore gives the same boundary sum as before contraction. In particular (15) continues to recover exactly the same normalized source row; no source occurrence is deleted. Contracting an edge of a tree leaves a tree. The remaining flows are still uniquely recovered by leaf elimination, or equivalently by composing the one-edge restoration above.

Finally, degrees and coefficient heights concern the polynomial after all monomial collisions, including equality and unit terms. Contracting vertices can create parallel prescribed incidences, so the initial simple graph argument alone is no longer a sufficient height certificate. The selection test reconstructs and collects the entire new energy and checks its actual integer coefficients. This establishes the stated height, exact degree and coordinate census only for the accepted finite instance. □

Corollary 4.4. *Any finite certified sequence of accepted internal-edge contractions preserves the calibrated integer zero set. Its inverse is obtained by restoring the deleted flows in reverse order. If all k deleted flows were distinct live coordinates and no other coordinate is removed, its independent-variable count decreases by exactly k.*

Proof. Compose the polynomial deletion and restoration maps of Theorem 4.3. Every step preserves the block boundary identity and leaves internal blocks as trees. The final independently collected array fixes the actual coordinate count and all measured polynomial invariants. □

This argument does not predict which edges pass the height test or minimize the support. A contraction can save a variable while creating more monomials; it can also fail the height test entirely. Adaptive selection is a construction method, not a proof of optimality. The released certificate freezes each accepted edge, so reproduction requires no heuristic search.

4.9 Unit homogenized equality cycles

The new support reduction generalizes the anchored three-cycle without adding a coordinate. Let a, b be the two aliases of one non-unit logical class, and let $u_1, \dots, u_\ell$ be existing unit aliases, already anchored by the unchanged unit energy U. Choose three expressions f_1, f_2, f_3, each equal to a, b, au_j or bu_j, with both underlying aliases a and b represented. Replace that class's equality penalty by

$$C(f_1, f_2, f_3) = f_1^2 + f_2^2 + f_3^2 - f_1 f_2 - f_2 f_3 - f_3 f_1. \tag{19}$$

No other incidence label or unit relation is changed.

Theorem 4.5. *The replacement preserves the calibrated integer zero set and the universal source lift. It introduces no variable and has degree at most four. Any frozen instance whose complete coefficient collection has height one and exact degree four is a height-one quartic on the same independent coordinate set.*

Proof. The polynomial is nonnegative for every real tuple because

$$2C(f_1, f_2, f_3) = (f_1 - f_2)^2 + (f_2 - f_3)^2 + (f_3 - f_1)^2.$$

At a zero of the complete energy, the separately nonnegative unit summand gives $u_j = 1$ for every unit alias. The three expressions then reduce to copies of a and b, with both present. Their equality is therefore exactly $a = b$. Conversely $a = b$ and all $u_j = 1$ make the new penalty vanish. All other summands and the source-block divergence identity are unchanged. The old and new energies have literally the same zero set in the same coordinate space, so this step has the identity map as both witness maps. The same source projection and unique tree-flow lift also remain inverse on zeros.

On the canonical lift of an arbitrary source tuple, all units are one and all aliases equal their logical values. Thus the new penalty vanishes identically there, not merely over source zeros. This proves that (17) is unchanged. Each f_i has degree at most two, so the new penalty has degree at most four.

These facts do not by themselves prove height one or decreased support. A product au_j can create monomials already present elsewhere. The complete energy is therefore collected after each proposed change, including incidence, other equality penalties and the unit path. The accepted frozen certificate must pass exact checks of every coefficient, degree, constant and live coordinate. No choice of f_i is assumed universally safe for height. □

This changes the polynomial away from its zeros, but retains the same source witnesses. The tested cycles save support through actual signed monomial collisions in the full energy, not through an uncollected count of terms in one penalty. The finite selection procedure is not an exhaustive search over height-one equations.

4.10 Six matched source improvements

Endpoint	Variables	Prior terms	New terms	Reduction
`minimum503`	503	24193	24183	10
`compact505`	505	20491	20405	86
`compact512`	512	18255	18184	71
`balanced557`	557	11999	11991	8
`sparse617`	617	8318	8314	4
`sparser713`	713	7176	7172	4

All six have exact degree four, coefficient height one, constant coefficient one, no equality dummy coordinates, and every declared unrestricted coordinate present. Each new point strictly improves support over its matched Round 20 arithmetic source and dimension, with height unchanged. These small support gains are not described as new dimension records.

The first source is the retained 241-variable shared-error source; the next two use the separate 242-variable power-base source. The last three retain the grouped 263-variable or sparse 285-variable sources. Their original source bindings and affine refactorizations are unchanged. The refined 505- and 512-variable points use the new split-factor normalization, with 342 normalized coordinates instead of 343; their final graph counts remain 505 and 512. The other four retain their constructor chains. In particular none is silently relabeled as the new pooled principal. Trials of a transfer to newer arithmetic sources did not give a selected improvement and are not counted as delivered equations.

To isolate the cycle change from normalization, the frozen cycle-only controls at 505 and 512 variables have 20483 and 18247 terms. Together with the other four endpoints these give the six matched cycle-only reductions $10, 8, 8, 8, 4, 4$. The refined selected graphs further save 78 and 63 terms against those two controls. Controls are replay evidence, not extra selected deliverables.

The release contains each literal full coefficient array, graph certificate, explicit source binding and frozen operation sequence. Independent reconstruction checks the new cycle identity and universal source lift as well as every expanded coefficient. The variable/support tradeoffs among these six selected outputs remain distinct; no exhaustive Pareto frontier or graph optimum is claimed.

5 Linear trace and inverse correction in ten positive integers

The three equations in this section have exactly ten independent inputs

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Their complete four-port BFS943/radix67 coefficient tuple is distinct from the six-port tuple of Section 6. Strict Horizons supplies the free quotient and projection-first strategy; Trace Radical supplies the linear trace; Unsquared Factors and Linear Selectors supplies the inverse correction at k. The new uniform signed bound licenses their composition without assuming $q \mid Y$ early and without the extra positive shift $T + \alpha$.

5.1 The three complete polynomial definitions

Put $h = \beta J$, $q = h + 1$, $C = q - F - \alpha$ and $k = \eta + \zeta$. Choose one of the following fixed charts. The symbols W_b, λ and D below are polynomial abbreviations, not independent inputs.

Chart	W_b	X	Y	λ	D
Free plain	qw	q^2w	s	q	q^7
Scaled inverse	w	qw	qs	1	$6q^6$
Free inverse	w	qw	s	q	$6q^7$

In every chart define

$$\begin{aligned}
c &= kY + \eta, & A &= \lambda Y(X+1) + 1, & \Delta &= A^2 - 1,\\
H &= 4A - 5, & v &= X\lambda^2 Y^2, & E &= 1 + v(v+1)k^2,\\
R &= [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J,\\
B &= (K_0 + W_b)C - F, & U &= ic^2, & a &= \Delta U^2 E,\\
F_0 &= 1 + a, & G &= EF_0, & f &= 1 + 2a,\\
Q &= 4\Delta a F_0, & b &= cT - R, & V &= fb - c,\\
K &= X(3c - 2X) + 2, & L &= bV, & M &= hDH.
\end{aligned}$$

In the plain chart put $e_c = k - R - 1 - q^4 s$; in either inverse chart put $e_c = 6(k - R - 1 - q^4 s) - vk(k^2 - 4)$. Finally set

$$\begin{aligned} Z &= hD[H(n - LB) + K] - H[DB + he_c], \\ P &= Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2). \end{aligned} \tag{20}$$

The equation is $P = 0$. Fixed powers are implemented by binary multiplications. There is no radical or quotient gate. The graph's register sharing, not the typography of these abbreviations, defines the measured operation count.

For the proof only, let $t_* = B/h$, $\kappa = K/H$, $z = Z/M$ and $W_* = 1 + hL$. In the plain chart let $e = k - R - 1$, $\rho = e/q^7$. In the inverse charts let

$$e_* = k - R - 1 - \frac{vk(k^2 - 4)}{6}, \qquad \rho = e_*/q^d, \quad d = 6 \text{ or } 7.$$

In all charts put $\sigma = Y/q^3$. Direct expansion gives

$$P = (Q - 1)Z^2 - M^2 G(QV^2 - 1), \qquad z = n + \kappa + \sigma - W_* t_* - \rho. \tag{21}$$

These proof quotients are rational until integrality is established. The carrier denominator is nonzero for every positive input.

5.2 Root extraction before any transport sign

Positive inputs give $q \geq 32$, $X \geq q$, $Y \geq 1$, $k \geq 2$, $c \geq 3$, $A > q^2$, $\Delta > 1$, $U \geq c^2$ and $E > 1$. The exact identities are

$$f^2 = 1 + 4aF_0, \quad Q = \Delta(f^2 - 1) = 4\Delta^2 U^2 G, \quad \gcd(E, F_0) = 1. \tag{22}$$

Also $f > 2c$ and $Q > c^2$, hence $V \neq 0$ for every integral b. At a zero, valuations in (21) imply that $G = g^2$ for a positive integer g: every prime of G divides neither $Q - 1$ nor $QV^2 - 1$, so its valuation in the rational square z^2 is even. Coprimality then gives $E = \tau^2$, $F_0 = f_0^2$ and $g = \tau f_0$ with positive integer roots.

The inherited adjacent-coefficient descent makes $y = |z|/g$ integral. For completeness, a reduced denominator m satisfies $m^2 \mid Q - 1$. For $u = |V| > 1$, the norm-preserving transformation

$$u' = (2Q - 1)u - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qu$$

has a nonzero integral first coordinate with $|u'| < u$ and preserves the reduced denominator of the second. The first integrality follows from $(Q - 1)y \in \mathbb{Z}$; denominator preservation follows from $\gcd(2Q - 1, m) = 1$. The norm gives $u < y < (2Q - 1)u/[2(Q - 1)]$, proving $0 < u' < u$ after reflection. Descent reaches $u = y = 1$, forcing $m = 1$. Therefore

$$QV^2 - (Q - 1)y^2 = 1, \qquad z = \varepsilon g y, \quad \varepsilon \in \{-1, 1\}.$$

With $I = 2Ug$, equations (22) give $f^2 - \Delta I^2 = 1$, $Q = (\Delta I)^2$, $c^2 \mid I$ and $c \leq I$. The retained signed all-index folding theorem applies to $V \equiv -c \pmod f$ and gives $c = \psi_A(p)$ for a positive odd index p. Since $c \geq 3$, $p \geq 3$, and hence

$$c \geq c_3 := 4A^2 - 1 = 4\Delta + 3 > 4q^4, \qquad c > X, \qquad K > 0. \tag{23}$$

No sign of R, B, C, b, V or divisibility $q \mid Y$ was used.

5.3 A uniform bound removes the positive shift

If $b = 0$, the auxiliary norm would give $0 < y^2 - c^2 = (c^2 - 1)/(Q - 1) < 1$, impossible for integers. Thus $b \neq 0$, $\operatorname{sgn} V = \operatorname{sgn} b$ and $|V| > 1$. Put $S = \Delta I$, so $Q = S^2$. Pell classification of (SV, y) gives an odd auxiliary index: even traces are units modulo S, whereas SV is divisible by S. Index one would give $|V| = 1$. The index is therefore at least three, and $|V| \geq 4Q - 3$. Moreover

$$\frac{a^2}{g^2} = \frac{\Delta U^2 a}{1 + a} > \frac{\Delta U^2}{2} > 9,$$

so $f > 6g$. Using $c_3 \leq c$ and $f > 2c$ yields the essential bound, for either sign of V,

$$|b| \geq \frac{|V| - c}{f} \geq 4\Delta f - \frac{c + c_3}{f} > 4\Delta f - 1 > 4g. \tag{24}$$

The possible auxiliary index three is harmless; it is not excluded. The norm also gives $y < 3|V|/2$. Consequently

$$L = bV = |b||V| > 0, \qquad gy < 3L/8 < L/2, \qquad |b|/q^d < L/2 \quad (d = 6, 7).$$

This remains valid at $Y = 1$, before its cubic q-scale is recovered.

In the plain chart $e = k - R - 1 < b$ since $cT - k + 1 > 0$. In the inverse charts $k \geq 2$ gives $e_* \leq e < b$. Thus $\rho < |b|/q^d$ in all charts. If $B \leq -1$, then $W_* t_* < -L$, and (21) implies

$$n = \varepsilon g y - \kappa - \sigma + W_* t_* + \rho < L/2 - L + |b|/q^d < 0,$$

contradicting the positive domain. Hence $B \geq 0$ and $C > 0$. This handles all negative values of R and zero uniformly. Only now use the authentic whole-tuple mask bound $3q < R < q^4 - 2$. Equation (23) gives $c > 2R$, hence $b, V > 0$. The retained normalized strong-rank and signed auxiliary-index theorems apply to $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$; the latter uses $f - 1 = 2\Delta i^2 c^4 E$. With $c > 2p, 2R$ they yield $p = R$. Thus R is positive odd, and mask parity gives even q and odd J. Neither ρ nor σ has yet been assumed integral.

5.4 Authenticate the first index at every prime

The square $E = \tau^2$ gives $k = 2\psi_{1+2v}(m)$ with $m \geq 1$. The case $m = 1$ would give $k = 2$, $\eta = \zeta = 1$ and $c = 2Y + 1$ strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. Thus $m \geq 2$. The first base $1 + 2v$ exceeds A, since $\lambda Y(2X\lambda Y - X - 1) > 0$. Monotonicity and $c > k$ then give $m < R$.

In the free plain chart $v = q^4 w s^2$. Integral z and $\gcd(q, hH) = 1$ give $q^7 \mid e - q^4 s$, hence $q^4 \mid e$. The recurrence $k \equiv 2m \pmod{4v}$ shows $q^4 \mid 2m - R - 1$. Its absolute value is less than $R < q^4$; therefore $2m = R + 1$.

In either inverse chart $v = q^3 w s^2$. The inherited finite Lucas identity, valid at every index, is

$$k = 2m + 8vC_m + 32v^2 Z_m, \quad C_m = \binom{m+1}{3}, \quad Z_m \in \mathbb{N}.$$

Since k is even, $k(k^2 - 4)/6$ is integral: write $k = 2u$ and use $3 \mid u(u - 1)(u + 1)$. Thus e_* is integral. Clearing the carrier and cancelling its literal factor six, not inverting six modulo q, gives $q^d \mid hH(e_* - q^4 s)$, hence $q^4 \mid e_*$. Put $t_0 = 2m$, $d_0 = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + kt_0 + t_0^2 - 4$. Cubic subtraction gives exactly

$$3(e_* - d_0) = 4v^2(24Z_m - A_m S_m). \tag{25}$$

Indeed $t_0(t_0^2 - 4) = 48C_m$ and $k - t_0 = 8vA_m$. For any prime $p \mid q$, writing $a_p = v_p(q) \geq 1$, the identity gives

$$v_p(e_* - d_0) \geq 6a_p - v_p(3) \geq 4a_p.$$

This includes the prime three. Hence $q^4 \mid d_0$, and the strict size bound gives $d_0 = 0$. In all three charts,

$$m = (R+1)/2. \tag{26}$$

Full q^6- or q^7-divisibility of e_* has not been inferred here; the third-primary obstruction is repaired only after projection.

5.5 Projection precedes complete denominator recovery

The literal numerator and integral z give $H \mid hDK$. Cancel only the q-part: $H \mid d_1 hK$, where $d_1 = 1$ in the plain chart and $d_1 = 6$ in the inverse charts. Write $R = 2r+1$, so $r \geq 48$. The retained λ-uniform ratio theorem applies to the authenticated indices and strip $Y < c/k < Y+1$:

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

Modulo odd H, the universal scaled recurrence gives $3 \cdot 2^{R-1}\psi_A(R) \equiv 4^R - 1$. The retained projector yields

$$H \mid d_1 h(X - 2^R)(2^R X + 1).$$

The complete product, including d_1, has magnitude below H:

$$d_1 h|(X - 2^R)(2^R X + 1)| < 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H.$$

After division by X^{r+1} the two terms decrease for $X \geq 32$, $r \geq 48$; at $(32, 48)$ their sum is $2^{-117} + 2^{-25} < 1$. Thus $X = 2^R$. The universal projector itself then gives positive integral $\kappa = K/H$; no unproved cancellation of $d_1 h$ is used.

Now $q \mid X$ implies $q = 2^a$ with $a \geq 5$. Since $R > 3q \geq 6a$, one has $q^6 \mid X$. In the free charts $v = Xq^2Y^2$ is divisible by q^8. In the plain chart $4v \mid e$ gives $q^7 \mid e$; in the free inverse chart (25) with $d_0 = 0$ gives $q^7 \mid e_*$. In the scaled chart $Y = qs$, so $q^8 \mid v$ as well and the same identity gives $q^6 \mid e_*$. Thus ρ is integral everywhere.

The carrier now makes $\sigma - W_* B/h$ integral. Multiply by $q^3 h$ and reduce modulo q^3 to obtain $q^3 \mid Y$; reduce modulo h to obtain $h \mid B$, using $\gcd(q, h) = 1$ and $W_* \equiv 1 \pmod{h}$. Thus $t_* = B/h \in \mathbb{N}$. These are the exact retained native outer conditions. The fixed-minus parity and half-binomial theorems, uniform in λ, reconstruct fresh source witnesses at $A_{\text{source}} = Y(X+1)+2$, not at the present base. The retained whole-tuple native/refutation calibration then supplies a finite syntactic ZFC refutation.

5.6 A constructive positive lift with possibly negative correction

Conversely a finite refutation supplies canonical outer data $q = 2^a$, $X = 2^R$, $R > 3q$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, $C > 0$ and integral $t_* = B/h \geq 0$. Choose $w = X/q^2$ in the free plain chart and $w = X/q$ otherwise; take $s = Y$ in free charts and $s = Y/q$ in the scaled chart. All are positive integers. Construct fresh A, v and set

$$c = \psi_A(R), \quad k = 2\psi_{1+2v}((R+1)/2), \quad \tau = \chi_{1+2v}((R+1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The retained exact uniform ratio theorem gives $0 < \eta < k$. The same recurrence and power-of-two divisibility prove integral ρ; projection gives $0 < \kappa < Xc$, and σ is positive integral.

Choose a positive Pell matrix period at A modulo $c^2\tau$, obtaining a seed (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$. Then $a = f_0^2 - 1$, $G = (\tau f_0)^2$ and the doubled strong pair is $f = 1 + 2a$, $I = 2f_0 I_0$. Keep this seed fixed. The retained positive auxiliary reconstruction gives (V_*, y) with $V_* = f^2(cT_{\text{old}} - R) - c$. Define

$$T_{\text{new}} = \frac{V_* + c + Rf}{cf}.$$

This is integral: $f^2 \mid V_* + c$, $V_* \equiv -R \pmod{c}$, $f \equiv 1 \pmod{c}$ and $\gcd(c, f) = 1$. It is strictly positive, and $V_* = f(cT_{\mathrm{new}} - R) - c$ exactly. This is a fresh witness construction, not an inverse shear on arbitrary positive tuples.

In inverse charts the correction may be negative. Since $k \geq R + 1$ and $vk^2 > 6$, one has $|\rho| < vk^3/3$. For every chosen $i \geq 1$,

$$g > \sqrt{\Delta}c^2E > c^2v^2k^2 > vk^3, \qquad g > 4Xc.$$

For the latter inequality use $\sqrt{\Delta} > A - 1 > X$ and $cE > 4$. Thus $\kappa + \sigma < g/2$ and $|\rho| < g/3$. In the plain chart $\rho \geq 0$, and the same bound on g suffices. In all cases

$$n = gy + (1 + hL)t_* + \rho - \kappa - \sigma$$

is a strictly positive integer. Its carrier has $z = gy$, so the literal polynomial vanishes. Every one of the ten independent inputs has been constructed in the actual positive domain at one coherent tuple.

5.7 Attained degrees and exact circuit counts

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$. Complete homogeneous collection gives

$$\begin{aligned}
[P_{\mathrm{plain}}]_{240} &= -2^8 q_0^{112} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2, \\
[P_{\mathrm{scaled}}]_{232} &= -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2, \\
[P_{\mathrm{free}}]_{214} &= -9 \cdot 2^{10} q_0^{86} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2.
\end{aligned}$$

They have respectively $99, 202, 99$ collected monomials. In the free charts the leading part of b is $-q_0^3(q_0 - F)$; in the scaled chart the full degree tie $b_0 = q_0 B_0$ must be retained. Selecting $\eta^{32}F^2$ in the free cases gives nonzero coefficients $-2^8\beta^{112}$ and $-9 \cdot 2^{10}\beta^{86}$; selecting $\eta^{34}T^2$ in the scaled case gives $-9 \cdot 2^{10}\beta^{100}$. These prove attained degree at the authentic positive β.

In chart order, the degrees of (Q, M, G, V, Z) are $(82, 13, 52, 40, 60)$, $(80, 11, 50, 40, 57)$ and $(72, 12, 46, 36, 54)$. The unique dominant output term is $-QM^2GV^2$. The live binary-operation counts are respectively $82, 85, 87$. The independent checker verifies complete top forms of every register, not sampled lines. The scaled 232/85 point saves two operations against incoming Trace Radical's 232/87; the free 214/87 point saves eighteen degrees against that same incoming point. Neither comparison is a global bound or a claim about unrestricted integer variables.

6 Two shorter circuits in eleven unrestricted integers

The independent coordinates are exactly $(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}$, with the six authentic source34 ports specified in Section 2. This round saves one binary operation from each frozen Round 20 endpoint. The degree-298 polynomial is literally unchanged; the degree-300 polynomial is changed by an explicit integral shear. The earlier degree reduction from the Nearest Factorial incoming frame is credited to that source.

6.1 Complete formulas and distinct arithmetic frames

Define

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\
X &= qW, & Y &= v, & E &= XY, \\
\ell &= XY^2, & a &= E + Y, & A &= a + 2, \\
H &= 4a + 3, & \Delta &= a^2 + H = A^2 - 1, & & \\
F &= (K_0 + W)C - z(q - 1), & & & & \\
R &= [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J. & & & &
\end{aligned}$$

For the degree-298 frame use $k = R{+}1{+}h\ell$ and $b_* = 6$. For the degree-300 frame use $k = R{+}1{+}hE$ and $b_* = 8$. Then put

$$\begin{aligned}
\eta &= c - Yk, \\
D_{\mathrm{err}} &= (\mathrm{cone_A}F - q)^2 + (\mathrm{cone_B}C - q)^2, \\
B_0 &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
\mathcal{B} &= Y^2 B_0 - k^2 q^{b_*}, & D_t &= 1 + \ell(\ell+1)k^2, \\
P &= D_t[1 + (\mathcal{B} - 4)D_t s^2], & D_m &= 1 + \Delta c^2, \\
B &= rAD_m, & U &= rc^2 D_m, \\
I_2 &= c^4 B = Ac^2 U, & D_f &= 1 + \Delta I_2, \\
Q &= \Delta^2 I_2, & \delta &= Q - 1, \\
V &= D_f(c - R + 4AUT) - c, & C_y &= QV^2 - 1, \\
K &= X(3c - 2X) + 2, & M &= 2q^3, \\
D_c &= HM, & S &= srHM^5, \\
z_c &= D_c T + MK + H(v - q^3), \\
A_c &= (z_c^2 + 3D_c^2)(z_c^3 - 19D_c^3), \\
\gamma &= 8\Delta + 4, & \lambda &= 4(\gamma U - rc^4).
\end{aligned}$$

Use the center matching the selected frame:

$$\begin{aligned}
\Psi_{298} &= \lambda V + U^2 T\{rc^4[\gamma(c^2 - 2) - c^2] + 64A\Delta\}, \\
\Psi_{300} &= \lambda V + U^2 T\{(\gamma - 1)[rc^4(c^2 - 2) + 8A]\}.
\end{aligned} \tag{27}$$

Finally define the ordinary polynomial

$$\begin{aligned}
Z &= S(n + \Psi) - A_c, \\
p_a &= \delta Z^2, \quad p_b = \delta S^2 P, \quad p_c = 1024 S^2 B D_f C_y, \\
\mathcal{N} &= (p_a + p_b - p_c)^2 - 4p_a p_b.
\end{aligned} \tag{28}$$

The equation is $\mathcal{N} = 0$. There are no extra quantified registers, division gates or radical gates. In particular the exponent b_* is one of two fixed integers, not a variable exponent.

6.2 The complete new zero set proof

The Round 20 degree-298 center used $rc^4[(\gamma - 1)(c^2 - 2) - 2] + 64A\Delta$ in its brace. The elementary identity

$$(\gamma - 1)(c^2 - 2) - 2 = \gamma(c^2 - 2) - c^2$$

removes one subtraction without changing any coefficient of the complete polynomial. The Round 20 degree-300 brace was $rc^4(\gamma - 1)(c^2 - 2) + 64A\Delta$. Since $\gamma - 1 = 8\Delta + 3$, the new brace exceeds it by $24A$. Therefore

$$\Psi_{300}^{\mathrm{new}} - \Psi_{300}^{20} = 24AU^2T =: \epsilon.$$

The polynomial ϵ is integral and independent of n. Keeping the ten other coordinates and setting $n_{20} = n_{21} + \epsilon$ makes the old and new Z, hence their entire polynomials, equal. The inverse is $n_{21} = n_{20} - \epsilon$. This is a bijection of the full unrestricted-integer zero sets. No positive-domain shear or unproved arithmetic condition is used. The changed factorization removes one multiplication.

For clarity, the earlier center compression is retained but not claimed anew. Relative to the incoming calibrated centers it uses the exact identities $U = rc^2 + \Delta rc^4$ and $A^3 = A\Delta + A$. The center differences to those parents are

$$U^2T(16rc^2 - 64A), \qquad U^2T(16rc^2 - 40A - 14rc^4)$$

for the present degree-298 and degree-300 equations respectively. They too are integral n-shears. The full parent proofs and exact source pins remain in the package, so the new calibration reduces to actual old zeros in both directions, not merely to necessary Pell properties.

6.3 The inherited arithmetic boundary stays unchanged

The degree-298 frame retains both its first-index modulus $\ell = XY^2$ and its budget $\mathcal{B}_6 = Y^2B_0 - k^2q^6$. It does not delete a guard by assuming early $q^3 \mid Y$. The parent even-weight norm, coprime-product arguments and integral denominator descent recover the positive individual square roots, $S \neq 0$, $S \mid A_c$, $\mathcal{B}_6 \geq 12$, and $c^2 \mid I$. The budget gives $|Y| > |q|^2$ and the authentic packing interval $|q|(|q|^2 - 1) < R < |q|^4 - 2$. Hence $|\ell| > 4|q|^5 > 3R + 1$. Signed rank and root folding recover the main index and then $X = 2^R$ before the cubic scale is extracted. Only then is q a signed power of two and odd H coprime to the carrier's $2q^3$ factor.

Soundness restores the earlier B_0-only native source, not the stronger old q^8 budget. The native first-index coordinate is rebuilt as $h_{\mathrm{old}} = Yh$, giving the same k exactly. Fresh strong and auxiliary witnesses are regenerated with their required parity. In completeness, the guard is chosen first, then one sufficiently divisible strong pair is fixed, then the intersective-quintic residue and supported-modulus interpolation are chosen with that same pair. Neither pair reselection nor an assumed odd incoming multiplier is used. The present one-gate changes touch only the final center evaluation and add no new sign, scale or interpolation hypothesis.

6.4 Exact attained degrees and retained restricted floors

In the free ring on (A, c, r, T, R) with weights $(3, 1, 1, 1, 5)$, put

$$D_\Psi = \Psi^2 - 1024BD_fV^2, \qquad E_\Psi = (Q-1)D_\Psi - 1024BD_f(V^2-1).$$

Complete collection gives weights $80, 108$ for the degree-298 pair, and $82, 110$ for the degree-300 pair. The full leading forms are

$$\begin{aligned}
[D_{298}]_{80} &= 512A^{15}c^{24}r^5TR - 8192A^{17}c^{22}r^5T^2,\\
[E_{298}]_{108} &= 512A^{22}c^{30}r^6TR - 24576A^{24}c^{28}r^6T^2,\\
[D_{300}]_{82} &= 512A^{16}c^{26}r^6T^2,\\
[E_{300}]_{110} &= 512A^{23}c^{32}r^7T^2.
\end{aligned}$$

The independent checker collects every remainder coefficient, not just these forms. The extra $24AU^2T$ has weight 26, below the decisive center terms, and does not change these complete leaders.

For $W_0 = Sn - A_c$ the exact cancellation cut is

$$p_a + p_b - p_c = \delta W_0^2 + 2\delta S\Psi W_0 + \delta S^2 P + S^2 E_\Psi.$$

The degrees of S, W_0, δ, Ψ are $20, 35, 28, 54$. The first three terms have degree at most $98, 137, 122$ in the rebased frame, with 124 for the third in the old frame. The last term has degree 148 or 150. For the rebased frame, the inner square has degree 296 while $-4p_ap_b$ has degree 298. In the old frame both final contributions can have degree 300, but their leading T-degrees are four and two, so they do not cancel. Attained actual coefficients are

$$\begin{aligned}
[j^{136}w^{46}v^{50}z^6s^6r^{12}c^{40}T^2]\mathcal{N}_{298} &= 2^{272}3^{182}\beta^{136} > 0,\\
[j^{110}w^{50}v^{50}s^4r^{18}c^{64}T^4]\mathcal{N}_{300} &= 2^{256}3^{160}\beta^{110} > 0.
\end{aligned}$$

The two literal graphs have $29 + 23 + 94 = 146$ and $29 + 23 + 92 = 144$ live additions, subtractions and multiplications. The retained degree floors are 298 and 300 only within their respective fixed arithmetic frames with a constant nonzero radical weight and an integer polynomial center independent of n. They are not operation minima or bounds for other consistency-equation architectures.

7 Incoming credit and the scope of improvement

The comparison uses the seven newly inspected incoming packages and the frozen Round 20 delivery. The repository snapshot is

`231a9cc9351bec9b70e334826058ed5d449c9460`

with recorded commit time 25 October 2026, 23:12:49 UTC. The provenance ledger binds the retrieved members to that snapshot or their Library input identities. It does not infer a mathematical theorem from a commit hash.

Pooled Powers
The incoming 225-variable quartic is the strongest new direct dimension baseline. Its two-row common-base aggregate and signed square-frame exactness theorem are used and credited, not relabeled as discoveries of this round. The actual selected source is nevertheless the distinct Strict Horizons source.

Quadratic Projection 234
The common-Hessian integral projection and affine-consequence technique supply the algebraic module. Its separate cofinal two-square beta-spacing construction is valid but is not an additional saving added to the selected strict source. Only transformations actually rebuilt and checked on the selected arrays contribute to the new counts.

Affine Bootstraps and Affine Certificates
These provide the affine signed-radix and odd beta-multiplier ideas and their noncircular sign order. The selected transfer retains the strict source's actual square-atom frame, not the different smaller affine frame that yields only a weaker atom bound. The exact beta edit is reproved with the new pooled requests and wider remainder sphere.

Strict Horizons
This is the direct principal's arithmetic ancestor. Its strict local ranges, self-power $F = t^t$, linear CRT radius, parity-compatible cofinal construction and factorial-free calibration remain the source boundary. Its compact free-quotient chart also enters the positive scalar branch.

Trace Radical
The linear Pell trace replaces its square in the positive scalar certificate. The latest incoming degree-232/cost-87 point is the matched comparison for the new 232/85 and 214/87 points.

Unsquared Factors and Linear Selectors
The inverse-Lucas correction evaluated at k is composed with that trace and the free quotient. The new uniform signed bound is what licenses this composition before $q \mid Y$ and without the extra positive shift. The three-primary issue is handled before asserting full denominator recovery.

The six height-one graphs retain their respective Round 20 arithmetic sources, not a purported transfer of all new pooled savings. The 11-integer branch retains its own source34 tuple and source theorem. The direct source has no huge machine coefficient; the compact scalar circuits deliberately retain large fixed numerical payloads. These resources cannot be compared merely by counting independent variables.

The program already contains lower-variable constructions with different degrees, coefficient payloads and calibration routes. The present results are improvements of specified explicit families, not a project-wide minimum number of variables, a literature-wide priority claim or a canonical simplest equation.

8 Reproduction and evidence boundaries

The complete release is distributed as two ordinary, disjoint ZIP parts, each no larger than 4 MB. Both use the same top-level directory. Extract both archives into the same parent directory before verifying; neither part alone is a complete release. From the combined extracted root run

```
python3 -B verify_release.py
```

The integrated verifier checks the sealed inventory and executes its explicit verification plan in a disposable temporary copy. The option `-output ../receipt.json` stores a fresh receipt outside the release; `-integrity-only` checks byte inventory rather than mathematical reproduction. Optimized checks set `PYTHONOPTIMIZE=1`, so child Python processes inherit that mode. Python 3.10 or later and its standard library suffice for the finite mathematical checks; no repository checkout, network connection, external computer algebra system or proof assistant is required.

8.1 Complete definitions and deterministic rebuilds

The three selected direct equations are under `signed/principal217`, `signed/low217` and `signed/dimension216`. Each contains its entire residual array and collected quartic. Frozen certificates specify the semantic source edits, two common-base groups, every shear and row operation, monic restoration, Hessian direction and affine consequence. Reproduction replays those finite choices; it does not rerun an optimizer. The retained strict arithmetic source and calibrated factorial-free proof closure are included alongside the pooled and quadratic modules.

The height-one catalog contains each full coefficient array, graph, source binding and frozen equality-cycle replacement. Its matched Round 20 inputs are authenticated before their support differences are measured. The positive scalar graphs are `scalar/positive10/positive10_degree240_ops82.json`, `positive10_degree232_ops85.json` and `positive10_degree214_ops87.json` in that same directory. The integer graphs are `scalar/integer11/integer11_degree298_ops146.json` and `integer11_degree300_ops144.json` in that directory. Their respective four-port and six-port numerical tuples have complete compressed payloads, finite reconstruction recipes and independent byte/bit-length and digest bindings. Hashes are not substituted for missing numerical values.

The native mathematical report is rebuilt with

```
sh report/build_report.sh
```

This requires pdfLaTeX. It compiles three passes in a fresh temporary directory and rejects overfull boxes or unresolved references. `report/render_residuals.py` generates the principal appendix directly from the literal JSON. Its receipt binds both source and TeX hashes; an independently implemented parser checks every printed coefficient and monomial. PDF font and metadata bytes may differ across TeX installations; the literal arrays and coefficient payloads are the normative mathematical objects.

8.2 What the finite checks establish

- Independent intake checking authenticates all seven root inventories, recollected 99 incoming direct expansions, and binds the relevant Git blobs and extracted member bytes. These checks establish identity and finite algebra, not every unbounded theorem of an incoming packet.
- The signed checker reconstructs the exact source mutations and all 34 affine square atoms, authenticates group ports and canonical-pair wires, and replays every algebraic operation. Both restoration directions, unit conditions, permitted quadratic substitutions and every final quartic coefficient are checked. The independent full-chain receipt binds the actual 217- and 216-variable outputs.

- The height-one checker reconstructs all divergence and equality squares independently, checks integrality of their half-sum and every collected coefficient, and verifies source occurrence coverage, constructor pullbacks and the universal lift including the hub term. The changed cycle ledger uses only the two declared aliases and existing anchored units. Its exact full-polynomial difference is checked; coefficientwise equality of old and new polynomials is not falsely asserted.
- The positive scalar checker imports no producer. It verifies free-ring identities at every live gate, complete leading forms, the cubic inverse identity, the trace substitution and all four numeral reconstructions. Both alternative operator labels at all 254 arithmetic gates are rejected: 508 mutations.
- The integer scalar checks prove the exact identical-center rewrite and integral center shear, collect both complete cancellation remainders, and establish attained actual degrees. A separate checker binds 23 full highest forms, including the rebased h terms. Both alternative operator labels at all 290 arithmetic gates are rejected: 580 mutations. Normal and optimized checks remain active with Python assertions disabled.

The release receipts state the exact commands executed and their terminal outcomes. Ordinary mathematical reviews are AI-assisted reviews, not represented as external human refereeing. Source hashes, a successful circuit replay and a finite corruption suite have different evidentiary roles and do not replace each other.

8.3 The unbounded and logical boundary

The new pooling exactness, signed bootstrap, positive transport and witness constructions are unbounded ordinary arguments, not extrapolated from finite tests. Their retained inputs include canonical Pell classification and rank/folding, prime-power CRT and uniform half-binomial results, and the identified arithmetic/native-to-refutation calibrations. The complete selected source texts, pins and notices remain in the release. A relocated historical path is resolved in its component closure map rather than silently changing its theorem or coefficient tuple.

The C222/ZF and ZF/ZFC interfaces remain those stated in Section 2. No entire transitive Lean or Rocq calibration build is claimed here. Inclusion of interface names is not a new formal proof object; finite verification does not prove the absence of zeros, consistency, or a global lower bound. These distinctions apply equally to the direct, height-one and scalar equations.

9 Further questions

9.1 Can algebraically related bases share one pool

The proved two-row module shares one base and one modulus $2An - n^2 - 1$. Can the requests at B and BH be pooled using their algebraic relation, without increasing residual degree or introducing more coordinates than are saved? A sum of errors from different moduli has no present exactness proof. This is not achieved in the delivered construction.

9.2 Can the next direction preserve quadratic degree integrally

The released reductions use common Hessian-kernel directions, unit slopes and affine consequences. General primitive directions could require a unimodular completion and a Bezout combination of slopes. Can a larger finite search find another certified quadratic-preserving projection, while controlling the collected height and support? Absence of one of the present pivot types is only a restriction on that search family, not a lower bound on all Diophantine presentations.

9.3 Can equality cycles and flow contractions be selected jointly

A cycle change modifies the monomial collisions available to an internal flow contraction. A contraction can in turn make a previously rejected cycle safe. Joint optimization of the frozen graph, row signs and unit choices may give more than the small separate support savings here. Every proposed move needs its zero-set proof and full coefficient test; an uncollected count of negative terms is not a height certificate.

9.4 Can the positive scalar degree fall without losing its sign proof

The degree-214 point uses an unshifted $b = cT - R$ and a uniform bound on $|b|$ before R is known positive. Can a different carrier lower the remaining degree while preserving this bootstrap, the prime-three correction and the positive witness lift? A circuit identity alone is insufficient if the proposed coordinate replacement leaves the positive orthant. The fixed-frame degree floors for the eleven-integer branch place no bound on this different ten-positive construction.

9.5 Can an emitted array be bound to a kernel checked calibration

A focused formalization target is the actual two-row aggregate identity, the signed small-error theorem, the exact projected residual arrays and the graph-energy transfer. A complete transitive proof-assistant build could then bind those finite arrays to the retained syntactic refutation calibration. Successful coefficient replay and authenticated theorem-name source files do not by themselves supply that stronger endpoint.

A Every residual of the principal equation

The following is the complete finite definition of the principal polynomial. Its independent coordinates are $x_0, \ldots, x_{216} \in \mathbb{Z}$, and

$$P(x_0, \ldots, x_{216}) = \sum_{i=0}^{153} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$
\begin{aligned}
R_0 &= x_0 - x_0x_2 - x_2 + x_7 \\
R_1 &= -x_2x_5 - x_3 + x_6x_7 \\
R_2 &= -1 + x_1 - x_6x_7 \\
R_3 &= 1 + x_3 - x_4x_{18} + x_{16} \\
R_4 &= -1 + x_8 - x_8^2 - 2x_8x_9 - x_9 - x_9^2 \\
&\quad + x_{19} \\
R_5 &= -1 - 24x_8 - 12x_8^2 - 8x_8x_9 - 16x_8x_{19} - 6x_9 \\
&\quad - 8x_9x_{19} - 14x_{19} - 4x_{19}^2 + x_{20} \\
R_6 &= -2 - x_8 - x_8^2 - 4x_8x_{20} - 4x_{20} - 4x_{20}^2 \\
&\quad + x_{21} \\
R_7 &= 1 + 2x_{21} - 4x_{21}^2 + x_{22} \\
R_8 &= -2 - x_9^2 - 2x_9x_{10} - x_{10} - x_{10}^2 + x_{23} \\
R_9 &= -1 + x_8 - x_8^2 - 2x_8x_{10} - x_9 - x_{10}^2 \\
&\quad + x_{24} \\
R_{10} &= 15 + 6x_8 - 32x_8^2 + 4x_8x_9 - 18x_8x_{10} - 16x_8x_{19} \\
&\quad - 4x_8x_{23} - 32x_8x_{24} + 25x_9 + 4x_9^2 - 8x_9x_{10} + 8x_9x_{19} \\
&\quad + 8x_9x_{23} - 3x_{10} + 9x_{10}^2 - 16x_{10}x_{19} - 8x_{10}x_{23} - 16x_{10}x_{24} \\
&\quad + 12x_{19} - 8x_{19}x_{24} - x_{20} + 4x_{23}^2 - 21x_{24} - 4x_{24}^2 \\
&\quad + x_{25} - 2x_{43} \\
R_{11} &= 14x_8 - 2x_8^2 - 8x_8x_9 + 8x_8x_{10} - 8x_8x_{23} + 24x_9 \\
&\quad + x_9^2 + 34x_9x_{10} - 16x_9x_{23} - 21x_{10} - 35x_{10}^2 + 16x_{10}x_{23} \\
&\quad + 24x_{10}x_{25} + 15x_{23} - 8x_{23}^2 - 4x_{25}^2 + x_{26} \\
R_{12} &= 2 - 8x_8 - 4x_8^2 - 8x_8x_9 - 8x_8x_{10} + 8x_8x_{23} \\
&\quad + 8x_8x_{24} - 6x_9 + 8x_9x_{23} + 8x_9x_{24} - x_{10} + 8x_{10}x_{23} \\
&\quad + 8x_{10}x_{24} - x_{19} + 4x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 6x_{24} \\
&\quad - 4x_{24}^2 + x_{27} \\
R_{13} &= 1 - 4x_8 + 2x_9 - 4x_9^2 - 8x_9x_{10} - 8x_9x_{27} \\
&\quad + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{27} - 2x_{19} + 4x_{27} - 4x_{27}^2 \\
&\quad + x_{28} \\
R_{14} &= 39 - 30x_8 - 8x_8^2 - 10x_8x_9 - 20x_8x_{10} - 4x_8x_{23} \\
&\quad - 16x_8x_{24} - 22x_9 - 19x_9^2 - 22x_9x_{10} - 24x_9x_{23} + 4x_9x_{24} \\
&\quad - 27x_{10} - 19x_{10}^2 - 24x_{10}x_{23} - 16x_{10}x_{24} + 3x_{19} - 33x_{23} \\
&\quad - 4x_{23}^2 - 20x_{24} - 4x_{24}^2 + x_{29} \\
R_{15} &= 12 + 26x_8 + 26x_8^2 + 4x_8x_9 + 28x_8x_{10} + 8x_8x_{23} \\
&\quad + 32x_8x_{24} - 18x_9 - 16x_9x_{23} - 8x_9x_{24} + 36x_{10} - 18x_{10}^2 \\
&\quad + 16x_{10}x_{23} + 32x_{10}x_{24} - 24x_{10}x_{25} - 32x_{10}x_{29} - 6x_{19} - 8x_{23}^2 \\
&\quad + 30x_{24} + 8x_{24}^2 + 4x_{25}^2 - x_{26} - 2x_{29} - 4x_{29}^2 \\
&\quad + x_{30} + 4x_{43} \\
R_{16} &= -3x_8 - 47x_9 - 30x_9^2 + 12x_9x_{10} + 24x_9x_{19} + 24x_9x_{23} \\
&\quad + 5x_{10} + 6x_{10}^2 + 16x_{19} - 4x_{19}^2 - 8x_{19}x_{23} + 8x_{23} \\
&\quad - 4x_{23}^2 - x_{24} + x_{31} \\
R_{17} &= -3 - 6x_8 + 2x_9 - 2x_{10} - 2x_{24} - 4x_{31}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - 4x_{31}^2 + x_{32} \\
R_{18} &= 21x_8 - 8x_8^2 - 2x_8x_9 - 32x_8x_{19} + 36x_8x_{20} + 12x_9 \\
&\quad - 6x_9x_{10} - 24x_9x_{19} + 32x_9x_{20} - 15x_{10} - 3x_{10}^2 + 3x_{19} \\
&\quad - 12x_{19}^2 + 32x_{19}x_{20} - 9x_{20} - 28x_{20}^2 + 6x_{21} - 3x_{23} \\
&\quad - 3x_{24} + x_{33} + 12x_{36} - 12x_{36}^2 + 3x_{37} \\
R_{19} &= 1 + 2x_{33} - 4x_{33}^2 + x_{34} \\
R_{20} &= -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35} \\
R_{21} &= -6 - 8x_8 - 24x_8^2 + 16x_8x_9 - 32x_8x_{19} - 19x_9^2 \\
&\quad + 10x_9x_{10} + 32x_9x_{19} + x_{10}^2 - 7x_{19} - 16x_{19}^2 - 5x_{23} \\
&\quad + x_{24} + x_{36} + 2x_{71} \\
R_{22} &= -4 - 3x_8 - 7x_9 - 4x_{10} - 2x_{23} - x_{24} \\
&\quad + 4x_{36} - 4x_{36}^2 + x_{37} \\
R_{23} &= -6x_8 + 2x_9 + 2x_{10} - 4x_{10}^2 - 8x_{10}x_{37} - 2x_{24} \\
&\quad + 4x_{37} - 4x_{37}^2 + x_{38} \\
R_{24} &= -x_{10} - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39} \\
R_{25} &= -2 + 2x_8 - x_8^2 - 4x_8x_{39} - 16x_9 + 27x_9^2 \\
&\quad + 6x_9x_{10} + 24x_9x_{23} - 33x_{10} - 9x_{10}^2 + 24x_{10}x_{24} - 9x_{23} \\
&\quad + 8x_{23}^2 - 8x_{23}x_{24} + 22x_{24} - 4x_{24}^2 + 2x_{39} - 4x_{39}^2 \\
&\quad + x_{40} - 3x_{43} \\
R_{26} &= 4x_8 - 2x_8^2 - 8x_8x_{39} - 16x_{10}^2 - 16x_{10}x_{40} + 4x_{39} \\
&\quad - 8x_{39}^2 - 4x_{40}^2 + x_{41} \\
R_{27} &= -16x_8^2 + 24x_8x_{10} - 16x_8x_{19} + 16x_8x_{27} + x_{10} - 9x_{10}^2 \\
&\quad + 12x_{10}x_{19} - 12x_{10}x_{27} - 4x_{19}^2 + 8x_{19}x_{27} - 4x_{27}^2 + x_{42} \\
R_{28} &= -1 + 6x_9 - 4x_9^2 + 8x_9x_{10} - 8x_9x_{23} - 6x_{10} \\
&\quad - 4x_{10}^2 + 8x_{10}x_{23} + 6x_{23} - 4x_{23}^2 + x_{43} \\
R_{29} &= -1 - x_{10} - x_{10}^2 - 4x_{10}x_{43} - 4x_{43}^2 + x_{44} \\
R_{30} &= 1 + x_9 - x_9^2 - 4x_9x_{44} - 4x_{44}^2 + x_{45} \\
R_{31} &= -1 - 6x_{45} - 4x_{45}^2 + x_{46} \\
R_{32} &= -1 - x_{10} - x_{10}^2 - 4x_{10}x_{46} - 4x_{46}^2 + x_{47} \\
R_{33} &= 2 + x_9 - x_9^2 - 4x_9x_{47} - 4x_{47}^2 + x_{48} \\
R_{34} &= -2x_{42} - 4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49} \\
R_{35} &= -9 + 7x_8 - x_8^2 - 4x_8x_{49} + 12x_{49} - 4x_{49}^2 \\
&\quad + x_{50} \\
R_{36} &= -1 - 6x_{50} - 4x_{50}^2 + x_{51} \\
R_{37} &= -x_{10} - x_{10}^2 - 4x_{10}x_{51} - 4x_{51}^2 + x_{52} \\
R_{38} &= -x_8 - x_8^2 - 4x_8x_{52} - 4x_{52} - 4x_{52}^2 + x_{53} \\
R_{39} &= -10 - x_9 + x_{10} - x_{23} - 14x_{53} - 4x_{53}^2 \\
&\quad + x_{54} \\
R_{40} &= 1 - 2x_9 + 2x_{10} - 2x_{23} + 4x_{54} - 4x_{54}^2 \\
&\quad + x_{55} \\
R_{41} &= -10 + x_9 - x_9^2 - 4x_9x_{55} - 14x_{48} - 4x_{48}^2 \\
&\quad - 4x_{55}^2 + x_{56} \\
R_{42} &= 21 + 28x_{48} + 8x_{48}^2 - 2x_{56} - 4x_{56}^2 + x_{57} \\
R_{43} &= 2 + 15x_9 - 35x_9^2 + 14x_9x_{10} + 24x_9x_{19} - 4x_{10}x_{19} \\
&\quad - 5x_{19} - 4x_{19}^2 - x_{23} + x_{58} \\
R_{44} &= 1 + 6x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} \\
&\quad - 4x_{57}^2 + 4x_{58} - 4x_{58}^2 + x_{59} \\
R_{45} &= -2 - 2x_{10} + 2x_{10}^2 + 8x_{10}x_{57} + 8x_{57}^2 - 2x_{59} \\
&\quad - 4x_{59}^2 + x_{60} \\
R_{46} &= -2 - 4x_{60} - 4x_{60}^2 + x_{61} \\
R_{47} &= -x_3x_{13} + x_{62} \\
R_{48} &= x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}
\end{aligned}
$$

$$
\begin{aligned}
R_{49} &= 12x_{10}x_{62} + x_{12} - x_{13} - 2x_{26}x_{62} - x_{64} \\
R_{50} &= -2x_{28}x_{62} + 2x_{41}x_{62} - 2x_{62} - x_{65} + x_{70} \\
R_{51} &= -12x_{10}x_{62} + x_{12} - x_{13} - 2x_{30}x_{62} - x_{66} \\
R_{52} &= x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67} \\
R_{53} &= x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68} \\
R_{54} &= x_{12} - x_{13} - 2x_{38}x_{62} - x_{69} \\
R_{55} &= x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70} \\
R_{56} &= -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 \\
&\quad + x_{71} \\
R_{57} &= 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72} \\
R_{58} &= -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 \\
&\quad + x_{73} - x_{74}^2 \\
R_{59} &= -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{74} \\
R_{60} &= -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77} \\
R_{61} &= -1 + 2x_3 + 6x_3x_9 - 2x_3x_{19} + x_{75} \\
R_{62} &= -x_8x_{212} - x_{12} + x_{76} + x_{212} \\
R_{63} &= -4x_8x_{62} - 2x_9x_{62} + x_9x_{212} + x_{12} - x_{13} - 2x_{19}x_{62} \\
&\quad + x_{78} \\
R_{64} &= -x_{12}x_{63} + x_{13}x_{63} + x_{79} \\
R_{65} &= -x_{64}x_{79} + x_{80} \\
R_{66} &= -x_{65}x_{80} + x_{81} \\
R_{67} &= -x_{66}x_{81} + x_{82} \\
R_{68} &= -x_{67}x_{82} + x_{83} \\
R_{69} &= -x_{68}x_{83} + x_{84} \\
R_{70} &= -x_{69}x_{84} + x_{85} \\
R_{71} &= -x_{70}x_{85} + x_{86} \\
R_{72} &= -x_{77}x_{86} + x_{87} \\
R_{73} &= -x_{78}x_{87} + x_{88} \\
R_{74} &= -x_{88}^2 + x_{89} \\
R_{75} &= -x_2 - x_2x_{16} + x_{16} + x_{90} \\
R_{76} &= -x_{14}x_{90} + x_{91} \\
R_{77} &= -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 \\
&\quad - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{94} - x_{73}x_{89} - x_{91}^2 \\
&\quad - x_{92}^2 \\
R_{78} &= -x_2x_{15} + x_{15} - x_{15}x_{90} + x_{92} \\
R_{79} &= -5x_8 + 3x_9 - 4x_{19} - 2x_{22} - 2x_{26} - 2x_{28} \\
&\quad - 2x_{30} - 2x_{32} - 4x_{34} - 2x_{35} - 2x_{38} - 2x_{41} \\
&\quad - 4x_{60} - 2x_{61} + x_{93} \\
R_{80} &= -x_{97} - 4x_{97}x_{210} + x_{99} \\
R_{81} &= -x_4x_{120} + x_{95}x_{202} - x_{98} - x_{99}x_{120} \\
R_{82} &= x_{100}x_{202} - x_{101}x_{123} - x_{102} \\
R_{83} &= x_{103}x_{202} - x_{104}x_{123} - x_{105} \\
R_{84} &= x_{106}x_{202} - x_{107}x_{123} - x_{108} \\
R_{85} &= x_{109}x_{202} - x_{110}x_{123} - x_{111} \\
R_{86} &= x_{112}x_{203} - x_{114}x_{123} - x_{115} \\
R_{87} &= x_{116}x_{203} - x_{117}x_{123} - x_{118} \\
R_{88} &= -x_{17}x_{120} + x_{123} \\
R_{89} &= x_{119} - x_{121}x_{123} - x_{122} \\
R_{90} &= -x_{123}x_{125} + x_{124} - x_{126} \\
R_{91} &= -x_{123}x_{128} + x_{127}x_{202} - x_{129} \\
R_{92} &= -x_{123}x_{131} + x_{130}x_{202} - x_{132} \\
R_{93} &= x_{133} - 2x_{133}x_{135} + x_{136} \\
R_{94} &= -x_{133}x_{136} - x_{136} + x_{137} \\
R_{95} &= 2x_{134} + x_{134}^2 + x_{136}^2 - x_{137}^2 \\
R_{96} &= -x_{134}x_{139} - x_{139} + x_{140}
\end{aligned}
$$

$$
\begin{aligned}
R_{97} &= -1 + x_{138}^2 + 2x_{138}x_{140} + x_{139}^2 \\
R_{98} &= 1 - 2x_0 + x_0^2 - 2x_0x_{134} + x_{141} \\
R_{99} &= -x_0x_{139} + x_3 - x_{138} + x_{141}x_{142} \\
R_{100} &= 1 - 2x_4 + x_4^2 - 2x_4x_{134} + x_{143} \\
R_{101} &= -x_4x_{175} + x_{17} + x_{143}x_{144} - x_{174} \\
R_{102} &= -1 + 2x_{134}x_{186} - x_{145} + 2x_{186} - x_{186}^2 \\
R_{103} &= 1 + x_{145}x_{146} - x_{174} - x_{175}x_{186} + 4x_{210} \\
R_{104} &= -x_{134}x_{148} - x_{148} + x_{149} \\
R_{105} &= -1 + x_{147}^2 + 2x_{147}x_{149} + x_{148}^2 \\
R_{106} &= 1 + x_{134} - x_{134}x_{151} - x_{151} + x_{152} \\
R_{107} &= x_{150}^2 + 2x_{150}x_{152} - 2x_{151} + x_{151}^2 \\
R_{108} &= -x_{134}x_{154} - x_{154} + x_{155} \\
R_{109} &= -1 + x_{153}^2 + 2x_{153}x_{155} + x_{154}^2 \\
R_{110} &= -x_{134}x_{157} - x_{157} + x_{158} \\
R_{111} &= -1 + x_{156}^2 + 2x_{156}x_{158} + x_{157}^2 \\
R_{112} &= -x_{134}x_{160} - x_{160} + x_{161} \\
R_{113} &= -1 + x_{159}^2 + 2x_{159}x_{161} + x_{160}^2 \\
R_{114} &= -x_{134}x_{163} - x_{163} + x_{164} \\
R_{115} &= -1 + x_{162}^2 + 2x_{162}x_{164} + x_{163}^2 \\
R_{116} &= -x_{134}x_{166} - x_{166} + x_{167} \\
R_{117} &= -1 + x_{165}^2 + 2x_{165}x_{167} + x_{166}^2 \\
R_{118} &= -x_{134}x_{169} - x_{169} + x_{170} \\
R_{119} &= -1 + x_{168}^2 + 2x_{168}x_{170} + x_{169}^2 \\
R_{120} &= -x_{134}x_{172} - x_{172} + x_{173} \\
R_{121} &= -1 + x_{171}^2 + 2x_{171}x_{173} + x_{172}^2 \\
R_{122} &= 1 - x_{134}x_{175} - x_{175} + x_{176} \\
R_{123} &= -1 + 2x_{174} + x_{174}^2 + 2x_{174}x_{176} + x_{175}^2 \\
R_{124} &= -x_{134}x_{178} - x_{178} + x_{179} \\
R_{125} &= -1 + x_{177}^2 + 2x_{177}x_{179} + x_{178}^2 \\
R_{126} &= -x_{134}x_{181} - x_{181} + x_{182} \\
R_{127} &= -1 + x_{180}^2 + 2x_{180}x_{182} + x_{181}^2 \\
R_{128} &= -x_{134}x_{184} - x_{184} + x_{185} \\
R_{129} &= -1 + x_{183}^2 + 2x_{183}x_{185} + x_{184}^2 \\
R_{130} &= -8 - 2x_0^2 - 2x_0x_3 - x_1^2 - x_2^2 - 2x_3^2 \\
&\quad - x_4^2 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 - x_{11}^2 \\
&\quad - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 \\
&\quad - x_{93}^2 - x_{95}^2 - x_{96}^2 - x_{100}^2 - x_{103}^2 - x_{106}^2 \\
&\quad - x_{109}^2 - x_{112}^2 - x_{113}^2 - x_{116}^2 - x_{119}^2 - x_{120}^2 \\
&\quad - x_{124}^2 - x_{127}^2 - x_{130}^2 + x_{133} - x_{186}^2 - 24x_{210} \\
&\quad - 32x_{210}^2 \\
R_{131} &= -x_4 - x_4x_8 - x_4x_9 - x_4x_{10} - x_4x_{11} - x_4x_{12} \\
&\quad - x_4x_{13} - x_4x_{14} - x_4x_{15} - x_4x_{16} + x_{186} \\
R_{132} &= -16 + x_0 - x_{187}^2 \\
R_{133} &= -x_{134}x_{189} - x_{189} + x_{198} \\
R_{134} &= -1 + x_{188}^2 + x_{189}^2 - x_{198}^2 \\
R_{135} &= -x_{191} - x_{191}x_{192} + x_{199} \\
R_{136} &= -1 + x_{190}^2 + x_{191}^2 - x_{199}^2 \\
R_{137} &= -x_{184}^2 + x_{200} \\
R_{138} &= x_{189} - 2x_{194}x_{200} - x_{200} \\
R_{139} &= -4x_{189} - 8x_{189}x_{193} + x_{192} \\
R_{140} &= -x_{134} - x_{188}x_{195} + x_{192} \\
R_{141} &= -x_{183} - x_{185} - x_{188}x_{196} + x_{190} \\
R_{142} &= -x_{93} - 4x_{184}x_{197} + x_{191} \\
R_{143} &= 26 + 2x_0^2 + 2x_0x_3 - 2x_0x_{139} - 2x_0x_{148} + x_1^2
\end{aligned}
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\begin{aligned}
&- 2x_1x_{175} + x_2^2 - 2x_2x_{166} + x_3^2 - 2x_3x_{148} + x_8^2 \\
&- 2x_8x_{151} + x_9^2 - 2x_9x_{154} + x_{10}^2 - 2x_{10}x_{157} + x_{11}^2 \\
&- 2x_{11}x_{160} + x_{12}^2 - 2x_{12}x_{163} + x_{13}^2 - 2x_{13}x_{169} + x_{14}^2 \\
&- 2x_{14}x_{172} + x_{15}^2 - 2x_{15}x_{178} + x_{16}^2 - 2x_{16}x_{181} + x_{93}^2 \\
&- 2x_{93}x_{184} - x_{134}x_{201} - 2x_{138} - 2x_{140} + x_{140}^2 - 2x_{147} \\
&- 2x_{149} + x_{149}^2 - 2x_{150} + 2x_{151} - 2x_{152} + x_{152}^2 \\
&- 2x_{153} - 2x_{155} + x_{155}^2 - 2x_{156} - 2x_{158} + x_{158}^2 \\
&- 2x_{159} - 2x_{161} + x_{161}^2 - 2x_{162} - 2x_{164} + x_{164}^2 \\
&- 2x_{165} - 2x_{167} + x_{167}^2 - 2x_{168} - 2x_{170} + x_{170}^2 \\
&- 2x_{171} - 2x_{173} + x_{173}^2 - 2x_{174} + x_{176}^2 - 2x_{177} \\
&- 2x_{179} + x_{179}^2 - 2x_{180} - 2x_{182} + x_{182}^2 - 2x_{183} \\
&- 2x_{185} + x_{185}^2 \\
R_{144} &= -x_{96}x_{202} + x_{120} \\
R_{145} &= -x_{113}x_{203} + x_{120} \\
R_{146} &= -1 + x_2 - x_3 - 2x_3x_{211} \\
R_{147} &= -2 - x_{93}^2 - x_{139}^2 - x_{148}^2 + 2x_{151} - x_{151}^2 \\
&- x_{154}^2 - x_{157}^2 - x_{160}^2 - x_{163}^2 - x_{166}^2 - x_{169}^2 \\
&- x_{172}^2 - x_{175}^2 - x_{178}^2 - x_{181}^2 + x_{184}^2 - x_{204}^2 \\
&- x_{205}^2 - x_{206} - x_{206}^2 \\
R_{148} &= -1 - 2x_{98}^2 - 2x_{102}^2 - 2x_{105}^2 - 2x_{108}^2 - 2x_{111}^2 \\
&- 2x_{115}^2 - 2x_{118}^2 + x_{120}^2 - 2x_{122}^2 - 2x_{126}^2 - 2x_{129}^2 \\
&- 2x_{132}^2 - x_{207}^2 - x_{208}^2 - x_{209} - x_{209}^2 \\
R_{149} &= -x_3x_{12} + x_{212} \\
R_{150} &= -2x_{95}x_{148} + 2x_{100} - 2x_{100}x_{151} - 2x_{103}x_{154} - 2x_{106}x_{157} - 2x_{109}x_{160} \\
&- 2x_{112}x_{163} - 2x_{116}x_{169} - 2x_{119}x_{172} - 2x_{124}x_{178} - 2x_{127}x_{181} - 2x_{130}x_{184} \\
&- 2x_{134}x_{214} + 2x_{147}x_{148} + 2x_{148}x_{149} - 2x_{150} + 2x_{150}x_{151} + 2x_{151}x_{152} \\
&- 2x_{152} + 2x_{153}x_{154} + 2x_{154}x_{155} + 2x_{156}x_{157} + 2x_{157}x_{158} + 2x_{159}x_{160} \\
&+ 2x_{160}x_{161} + 2x_{162}x_{163} + 2x_{163}x_{164} + 2x_{168}x_{169} + 2x_{169}x_{170} + 2x_{171}x_{172} \\
&+ 2x_{172}x_{173} + 2x_{177}x_{178} + 2x_{178}x_{179} + 2x_{180}x_{181} + 2x_{181}x_{182} + 2x_{183}x_{184} \\
&+ 2x_{184}x_{185} + 4x_{210}x_{214} + x_{213} \\
R_{151} &= 1 + x_{93}^2 + x_{95}^2 - 2x_{95}x_{147} + x_{100}^2 - 2x_{100}x_{150} \\
&+ x_{103}^2 - 2x_{103}x_{153} + x_{106}^2 - 2x_{106}x_{156} + x_{109}^2 - 2x_{109}x_{159} \\
&+ x_{112}^2 - 2x_{112}x_{162} + x_{116}^2 - 2x_{116}x_{168} + x_{119}^2 - 2x_{119}x_{171} \\
&+ x_{124}^2 - 2x_{124}x_{177} + x_{127}^2 - 2x_{127}x_{180} + x_{130}^2 - 2x_{130}x_{183} \\
&+ x_{139}^2 + x_{147}^2 + x_{150}^2 + x_{153}^2 + x_{156}^2 + x_{159}^2 \\
&+ x_{162}^2 + x_{166}^2 + x_{168}^2 + x_{171}^2 + x_{175}^2 + x_{177}^2 \\
&+ x_{180}^2 + x_{183}^2 - 2x_{184}^2 + x_{204}^2 + x_{205}^2 + x_{206} \\
&+ x_{206}^2 - 4x_{210}x_{213} - 2x_{213} + x_{214} \\
R_{152} &= -2x_{96}x_{139} - 2x_{113}x_{166} - 2x_{120}x_{175} - 2x_{134}x_{216} + 2x_{138}x_{139} + 2x_{139}x_{140} \\
&+ 2x_{165}x_{166} + 2x_{166}x_{167} + 2x_{174}x_{175} + 2x_{175} + 2x_{175}x_{176} + 4x_{210}x_{216} \\
&+ x_{215} - x_{216} \\
R_{153} &= x_{96}^2 - 2x_{96}x_{138} + x_{113}^2 - 2x_{113}x_{165} + x_{120}^2 - 2x_{120}x_{174} \\
&+ x_{138}^2 - x_{139}^2 + x_{165}^2 - x_{166}^2 + x_{174}^2 - x_{175}^2 \\
&- 4x_{210}x_{215} - x_{215} + x_{216}
\end{aligned}
$$